Requirement Analysis and Specification Document
Students&Companies

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Contents

Chapter 1

Introduction

1.1 Purpose

As the demand for skilled professionals continues to grow, connecting university students with real-world opportunities has become increasingly important. Traditionally, students search for internships through fragmented processes, often relying on personal networks or generalized job boards, which may not adequately match their skills and aspirations with the needs of companies. This inefficiency results in missed opportunities for both students and employers.
Students&Companies (S&C) is a platform designed to address this challenge by streamlining the internship matchmaking process. It bridges the gap between students seeking internships and companies offering them. S&C allows students to showcase their experiences and skills through CVs while enabling companies to describe their projects and requirements in depth.
Additionally, S&C supports the selection process by facilitating interviews and feedback collection. With features for proactive searching and personalized suggestions, the platform aims to create a seamless and transparent experience for all parties involved. The main goals of the platform are the following:

1.1.1 Goals

[G1] Companies find suitable candidates based on the requirements of their projects and positions

[G2] Students find internships matching their profile and experiences

[G3] Students proactively look for internships that align with their skills, experiences, and preferences

[G4] Companies are facilitated in managing interviews

[G5] Companies manage student admissions to internships

[G6] Students provide feedback about the internships

[G7] Companies provide feedback about the internships

1.2 Scope

In this section, we will define S&C platform domain.
There are two primary users of the system: **Students** and **Companies**.
Students are the individuals seeking internship opportunities. They can create and manage their profiles by uploading their CVs. The platform will notify students of relevant internship opportunities based on their profile information and allow them to apply directly. In addition, students can track the status of their applications and receive feedback from companies. The system also provides a Recommendation Engine to suggest

internships tailored to students' qualifications and preferences. Students can actively participate by proactively searching for internships or passively by checking suggested opportunities.
Companies are the organizations offering internship opportunities. They will be able to post internship vacancies, specifying the required skills, tasks, technologies, and terms. Companies can manage and edit their internship postings, as well as review and evaluate student applications. The system will help companies identify students who best match their internship requirements through a recommendation system. Companies will also be able to schedule interviews, manage the selection process, and check and respond to feedback provided by students.

1.2.1 World Phenomena

[WP1] Students develop skills through experiences which end up on their CVs

[WP2] Companies define the available internships

[WP3] Students decide they want to partake in an internship

[WP4] Students conduct internships

[WP5] Students form their opinions about the internships

[WP6] Companies form their opinions about the internships

1.2.2 Shared Phenomena

World controlled

[SP1] Students upload CVs on the platform

[SP2] Students browse available internships

[SP3] Students submit applications for internships

[SP4] Students and companies set up an interview

[SP5] Students participate in interviews with the companies

[SP6] Companies conduct interviews with the students

[SP7] Students provide feedback about internships

[SP8] Companies create internships announcements

[SP9] Companies review students' profiles

[SP10] Companies reach out to students with matching CVs

[SP11] Companies review students' applications

[SP12] Companies accept a student to an internship

[SP13] Companies provide feedback about students

Machine controlled

[SP14] The system notifies students about internships matching their profiles

[SP15] The system notifies to companies about suitable student profiles

[SP16] The system generates internship recommendations for students

[SP17] The system generates student profile recommendations for companies

[SP18] The system prompts students to provide feedback

[SP19] The system prompts companies to provide feedback

[SP20] The system tracks the status of matchmaking processes

1.3 Definitions, Acronyms, Abbreviations

1.3.1 Definitions

- **Recommendation Engine**: System which manages the recommendation of students' CVs to companies and of internships to students

1.3.2 Acronyms

- S&C: Students&Companies

1.3.3 Abbreviations

- G*: goal
- WP*: world phenomena
- SP*: shared phenomena
- R*: functional requirement
- UC*: use case

1.4 Revision history

Revision	**Date**	**Author(s)**	**Description**
1.0	25.11.2024	L. Ilardi, M. Luraghi	Chapter 1, init Chapter 2
2.0	02.12.2024	L. Ilardi, M. Luraghi	finished Chapter 2, init Chapter 3 (except no 3.1)
3.0	09.12.2024	L. Ilardi, M. Luraghi	continued Chapter 3
4.0	16.12.2024	L. Ilardi, M. Luraghi	continued Chapter 3, finished 3.1
5.0	21.12.2024	L. Ilardi, M. Luraghi	finished Chapter 3, Chapter 4, Chapter 5

1.5 Reference Documents

The document is based on the following materials:

- The specifications RASD and DD assignment of the Software Engineering II course a.a. 2024/2025
- Course slides on Webeep

1.6 Document Structure

1. **Introduction**: it aims at describing the project briefly. In particular, it focuses on the reasons and the goals that are going to be achieved with its development
2. **Overall Description**: it is a high-level description of how the system works with a detailed explanation of the phenomena that involve the world, the machine, or both. It also contains the description of the domain with its assumptions
3. **Specific Requirements**: this section provides a detailed analysis of the requirements needed to achieve the goals. Moreover, it contains additional information useful for developers (i.e. constraints about HW and SW)
4. **Formal Analysis Using Alloy**: it's a formal description of the world controlled shared phenomena by the means of Alloy
5. **Effort Spent**: it shows the time spent to realize this document organized by section

Chapter 2

Overall Description

2.1 Product perspective

2.1.1 Scenarios

User signs up
User A opens the landing page of the S&C platform and clicks on the "Sign up" button. Two different registration forms appear to the user, one for companies and the other for students. User A selects and fills in the form corresponding to their role. When User A is done the form can be submitted, and if everything checks out, the system creates the User's account and redirects them to the homepage.

User logs in
User B opens the landing page of the S&C platform and clicks on the "Log in" button. User B inserts his username and password (whether they are a Student or a Company) and if the credentials are correct the system redirects User B to the platform homepage.

Student uploads CV
Student C wants to upload his CV, so they log into the S&C platform. From the homepage, they navigate to the "Profile" section, where Student C can drag and drop the file or choose it from the file system by clicking the "Browse files" button. After the file has been uploaded, they click the "Upload CV" button to submit the CV, which will now appear on their profile page.

Student actively searches for an internship, finds it and applies
Student D wants to look for available internships that match their skills and experiences, so they log into the S&C platform. On the homepage they are greeted with the list of all available internships, they browse them and then select the one which suits them the most. The system shows Student D the page containing the internship's details, they click on the "Apply" button to enrol. Consequently, the system prompts a confirmation message and redirects Student D to the homepage.

Student applies to an internship proposed by a company
Student E logs into the S&C's platform and from the homepage, they navigate to the "Company proposals" section and select the internships they want to check out, proposed by Company F to Student E. The system shows Student E the page containing the internship's details, and they click on the "Apply" button to enrol. Consequently, the system prompts a confirmation message and redirects Student E to the homepage.

Student applies to an internship recommended by the system
Student G wants to look for available internships that match their skills and experiences, so they log into the S&C platform. From the homepage, they navigate to the "Suggestions" section, where there's a list of internships recommended by the system based on Student G's skills and experiences, they browse them and then

select the one which suits them the most. The system shows Student G the page containing the internship's details, they click on the "Apply" button to enrol. Consequently, the system prompts a confirmation message and redirects Student G to the homepage.

Company creates an internship announcement
Company H wants to create an announcement for a newly available internship, so it logs into the S&C platform. From the homepage, it navigates to the "Internships" section, where it can click the "New" button and fill the form with all the necessary information. After completing the form, Company H clicks the "Submit" button to create the internship announcement.

Company modifies an existing internship announcement
Company I has changed some details regarding an existing internship and must now modify the announcement accordingly. It logs in to the S&C platform and navigates to the "Internships" section. It selects the internship that must be modified and clicks the "Modify" button. It can now insert new information or change the old details, and at last, confirm the changes by pressing the "Submit" button.

Company deletes an existing internship announcement
Company J has decided to remove an internship opportunity and must now delete the announcement on the platform. It logs in to the S&C platform and navigates to the "Internships" section. It selects the internship that must be deleted and clicks on the "Delete" button. A pop-up warning asks Company J for confirmation, and when the "Confirm" button is pressed the post is removed by the system.

Company reviews a student's application
Company K logs into the S&C platform to review student applications for an internship. From the homepage, it navigates to the "Internships" section, which displays a list of internship announcements posted by the company. Company K selects a specific internship by clicking on it, which opens a detailed view of the internship. Within the internship's page, Company K accesses the "Applications" section, where it finds Student L's application. The system displays the detailed application, including Student L's profile and CV, by clicking on the application. After reviewing the CV, Company K decides to either accept or reject the application by clicking either the "Accept" or the "Reject" buttons.

Company reaches out to student offering an internship
Company M logs into the S&C platform to identify a suitable candidate for one of its internships. From the homepage, Company M navigates to the "Internships" section and selects a specific internship. Within the internship's page, Company M accesses the "Suggestions" section, where the system provides a list of recommended students. After reviewing the suggestions, Company M selects the most suitable student and clicks the "Notify" button to let Student N know that Company M proposed an internship to them. If Student N accepts the offer, they are ready to schedule an interview, without the need from Company M to evaluate Student N's CV again.

Student and company set up the interview
After approving Student O's application, Company P navigates to the "Interviews" section of the S&C platform. Here, Company P browses the students who were accepted for an interview, and finds Student O's entry where it is possible to select and propose three possible dates and times for the interview through a calendar pop-up. Student O, upon logging into the platform, accesses the "Applications" section to view the proposed dates. Student O can either: accept one of the suggested dates by clicking on it, confirming the choice; request a new set of dates by clicking the "Reschedule" button, prompting Company N to provide alternative options.

Student and company meet for the interview
At the agreed-upon date and time, Company Q and Student R log into the S&C platform using their credentials. Both navigate to the "Interviews" section, where a link to the third-party meeting application is available. By clicking the link, they are redirected to the meeting platform and start the interview.

Company admits a student to an internship
Company S has decided to admit Student T to the internship they interviewed for. Company S logs in to the S&C platform and navigates to the "Internships" section. It selects the internship it wants to admit Student

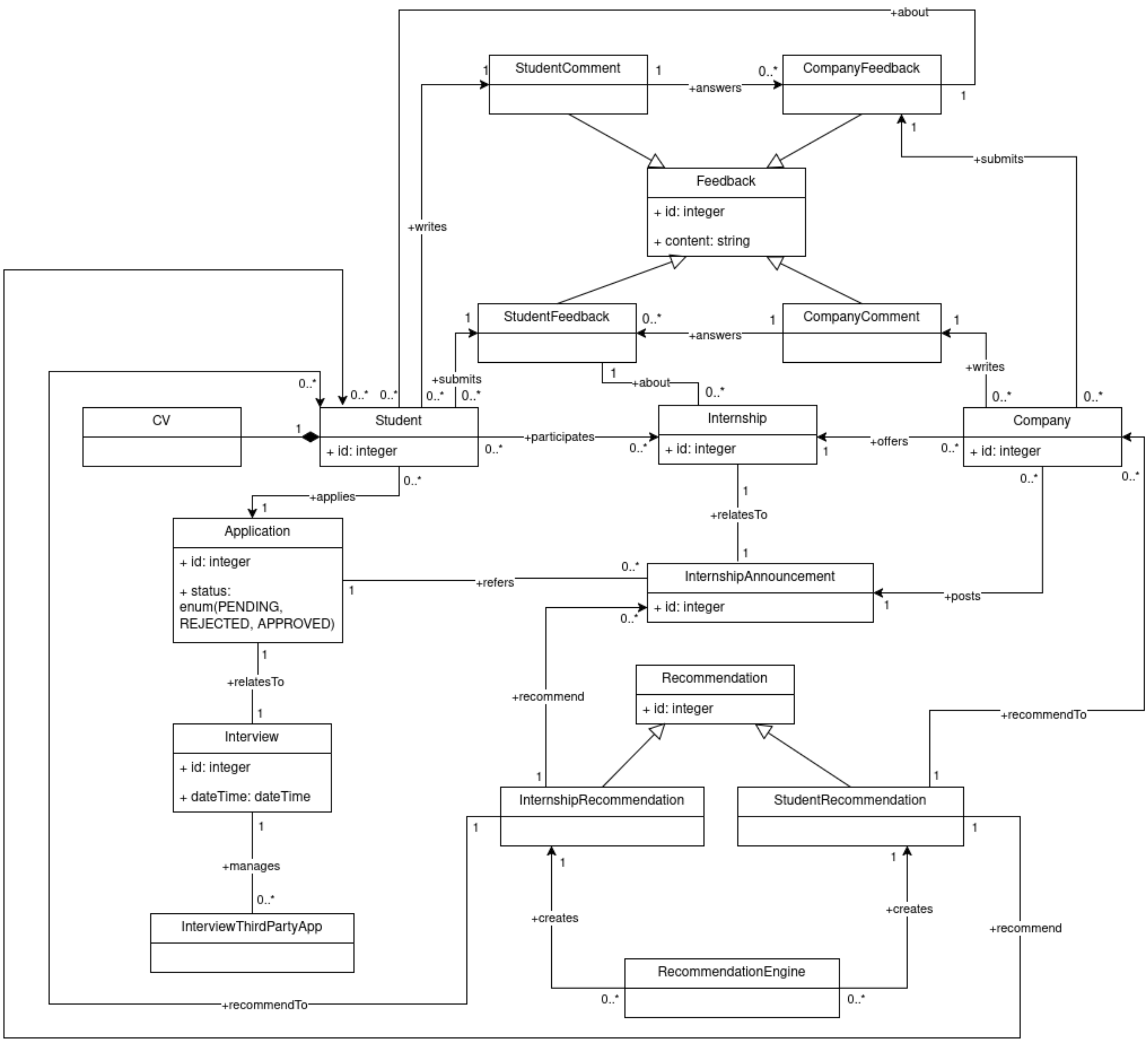
T to. Within the internship's page, Company S accesses the "Applications" section, where it finds Student T's approved application, it clicks on the "Select" button. A pop-up asks Company S for confirmation, and when the "Confirm" button is pressed Student T is admitted to the internship.

Student checks for an internship admission
Student U wants to check if they've been admitted to the internship offered by Company V. They log in to the S&C platform and navigate to the "Internships" section. Within the internship's page, Student U can see in the "Ongoing internships" section the list of internships they've been admitted to.

Student provides feedback on an internship
Student W logs into the S&C platform to provide feedback about an internship with Company X. From the homepage, Student W navigates to the "Internships" section and selects a specific internship. There's a comment section where Student W can provide feedback or respond to comments from Company X about Student W. After typing their comment, Student W clicks the "Submit" button, and the comment is displayed in the section.

Company provides feedback on a student
Company Y logs into the S&C platform to provide feedback about Student Z. From the homepage, Company Y navigates to the "Internships" section and selects a specific internship. There's a comment section where Company Y can provide feedback or respond to comments from Student Z about the internship with Company Y. After typing their comment, Company Y clicks the "Submit" button, and the comment is displayed in the section.

2.1.2 Domain Class Diagram



Domain Class Diagram Description

The Student class is associated with a single CV, which is the document used to apply for an internship. A Student submits their application in response to an Internship Announcement. Each application has a Status that is initially set to PENDING. After the company reviews the application, the Status transitions to either APPROVED or REJECTED. For applications marked as APPROVED, the system facilitates scheduling an interview between the student and the company through a third-party meeting application. After the interview, companies decide which students to admit to the internship, that become participants of the internship. Companies can offer multiple Internships, which are advertised through Internship Announcements. These announcements are the interface where students submit their applications. The system incorporates a Recommendation Engine that provides two types of recommendations:

1. Internship Recommendations: a list of internship opportunities proposed to students based on their profiles and preferences

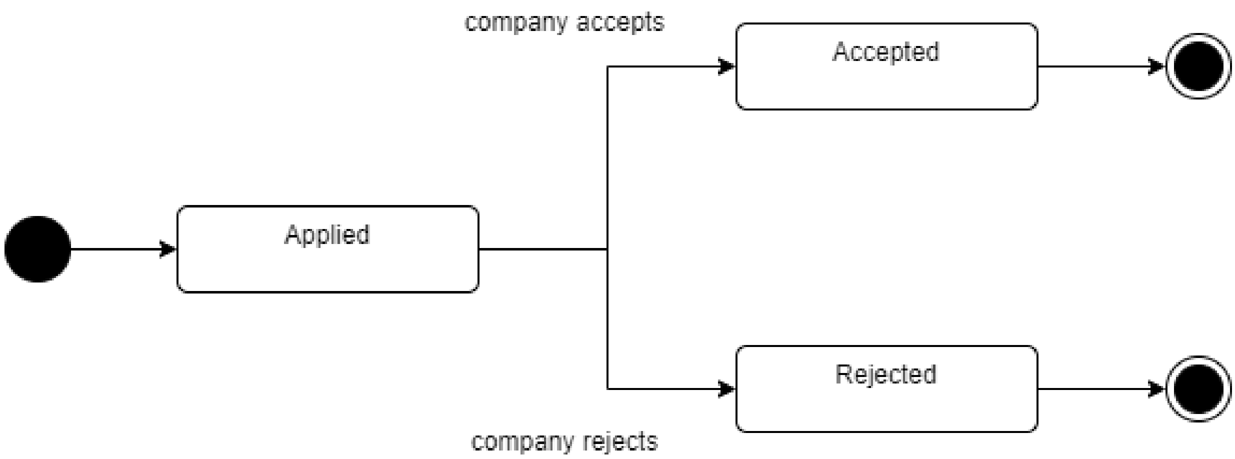
2. Student Recommendations: a selection of student profiles proposed to companies based on their requirements and preferences

The system also supports Feedback Mechanisms to enhance transparency and improve future recommendations:

1. Student Feedback: after completing an internship, students can provide feedback about the internship experience and the company
2. Company Feedback: companies can provide feedback on the performance and professionalism of students who have interned with them
3. Student Comment: students can respond to feedback provided by companies about them
4. Company Comment: companies can respond to students' feedback regarding their internships

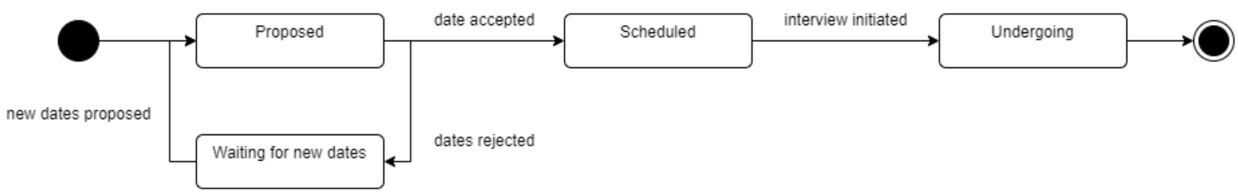
2.1.3 State Diagrams

Internship Application



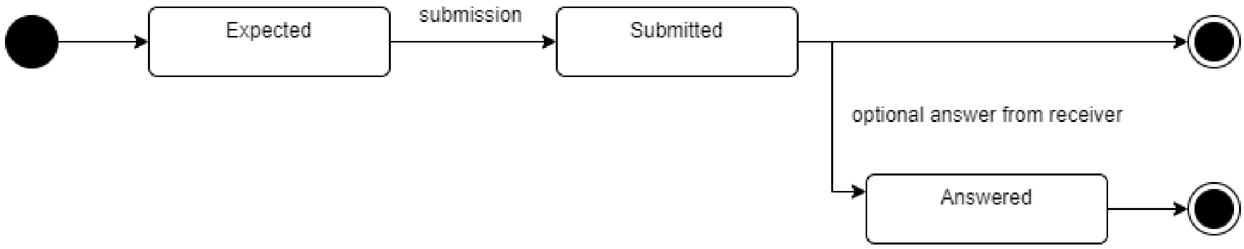
The Application exclusively refers to the process of preliminary admission to the interview. Because of this, applications are requests for an interview from students who either received a recommendation for a company from the system or actively searched for an internship. Students who got selected directly from a company thanks to a recommendation from the system to the company are automatically admitted for an interview since the company already reviewed the necessary information. When an Application is submitted, its status is Pending. The company proceeds to evaluate the candidates based on their CVs, and the Application can now be rejected or accepted. If accepted, the company initiates the process to schedule an interview with the student who applied.

Interview



An Interview is Proposed by the company to a student whose Application was accepted. The company proposes three possible dates for the interview. The student can either accept one of the suggested dates or refuse them, in which case the company will provide new dates. In the meantime, the status is Waiting for new dates. When the student accepts a date, the Interview is Scheduled, and on the given date it is Undergoing.

Feedback Submission



This diagram holds for both the students' and the companies' feedback. Feedback is expected from both, because of this the first state of the Feedback is Expected. When the feedback is Submitted it is now public for all to see. The party who received the feedback can now decide to optionally give a comment or answer to the statement, in which case it becomes Answered.

2.2 Product functions

Sign up and log in
The system allows users to register on the platform through a sign-up process. When a user clicks the "Sign Up" button, the system presents two distinct registration forms: one for students and the other for companies. The user must complete the form corresponding to their role by providing the required information. Once the form is successfully submitted, the system creates the user's account and redirects them to the homepage. After registration, users can access the platform by logging in with the credentials they provided during the sign-up process.

Internship announcement posting
The system allows companies to post detailed internship announcements. These postings include comprehensive information about the internship.

Internship announcement management
The system allows companies to manage existing internship announcements by modifying posts on the platform. Companies can decide to modify the details of the announcements or even delete them.

Internship application
The system enables students to apply for internships that align with their skills and experiences. Students can browse available opportunities. Once a suitable internship is identified, students can submit their application directly through the platform, allowing organizations to accept or reject their submissions.

Student recommendation
The system provides recommendations of students to companies by matching the students' skills and experiences with the requirements of available internships. Using the data provided by students by uploading their CVs, the system identifies suitable candidates for each internship and recommends them to the respective companies.

Company recommendation
The system recommends companies' internships to students by matching the students' CVs with the requirements of available internships.

Company reach out
The system allows companies to review students' profiles and CVs recommended by the system to identify

potential candidates for their internships. When a student's profile matches the requirements of a specific internship, the company can inform the student about an opportunity, inviting them to apply for the internship.

Interview hosting
The system enables companies and students to meet through an online interface for conducting interviews.

Student selection
The system allows companies to select students that have partook in an interview for an internship to be admitted for that specific internship.

Student feedback submission
The system collects feedback from students about their internship experiences. After completing an internship, students are asked to provide their opinions and evaluations through the platform. If the company wishes to, it can leave a comment regarding the student's feedback.

Company feedback submission
The system collects feedback from companies regarding the performance of students participating in their internships. After an internship concludes, companies are asked to submit evaluations through the platform, detailing the students' skills, professionalism, and overall contributions. If the student wishes to, they can leave a public comment or answer the company's feedback.

2.3 User characteristics

Two types of users interact with the platform: Students and Companies

2.3.1 Student

Students can create an account and log in to access the platform, where they can actively search for and apply to internship opportunities or wait to receive personalized suggestions for internships that align with their profile. They can also coordinate with companies to schedule interviews through the platform. After completing an internship, students have the opportunity to provide feedback about their experience.

2.3.2 Company

Companies can create an account and log in to access the platform, where they can post detailed internship announcements for students to view and apply to. They can review student applications, approve or reject them as needed, and inform students about internships that match their profiles. Companies can also coordinate with students to schedule interviews, admit students to an internship after an interview and provide feedback on the student's performance after the completion of an internship.

2.4 Assumptions, dependencies and constraints

2.4.1 Domain Assumptions

The following domain assumptions are properties or conditions that the system will take for granted, they must be checked to ensure a correct platform behaviour:

[D1] Users have a reliable Internet connection

[D2] Students provide accurate and up-to-date CVs

[D3] Companies properly insert information about internships in the announcements

[D4] Both parties attend the scheduled interviews on the platform

[D5] Students carry out the internships offered by the companies they were chosen for

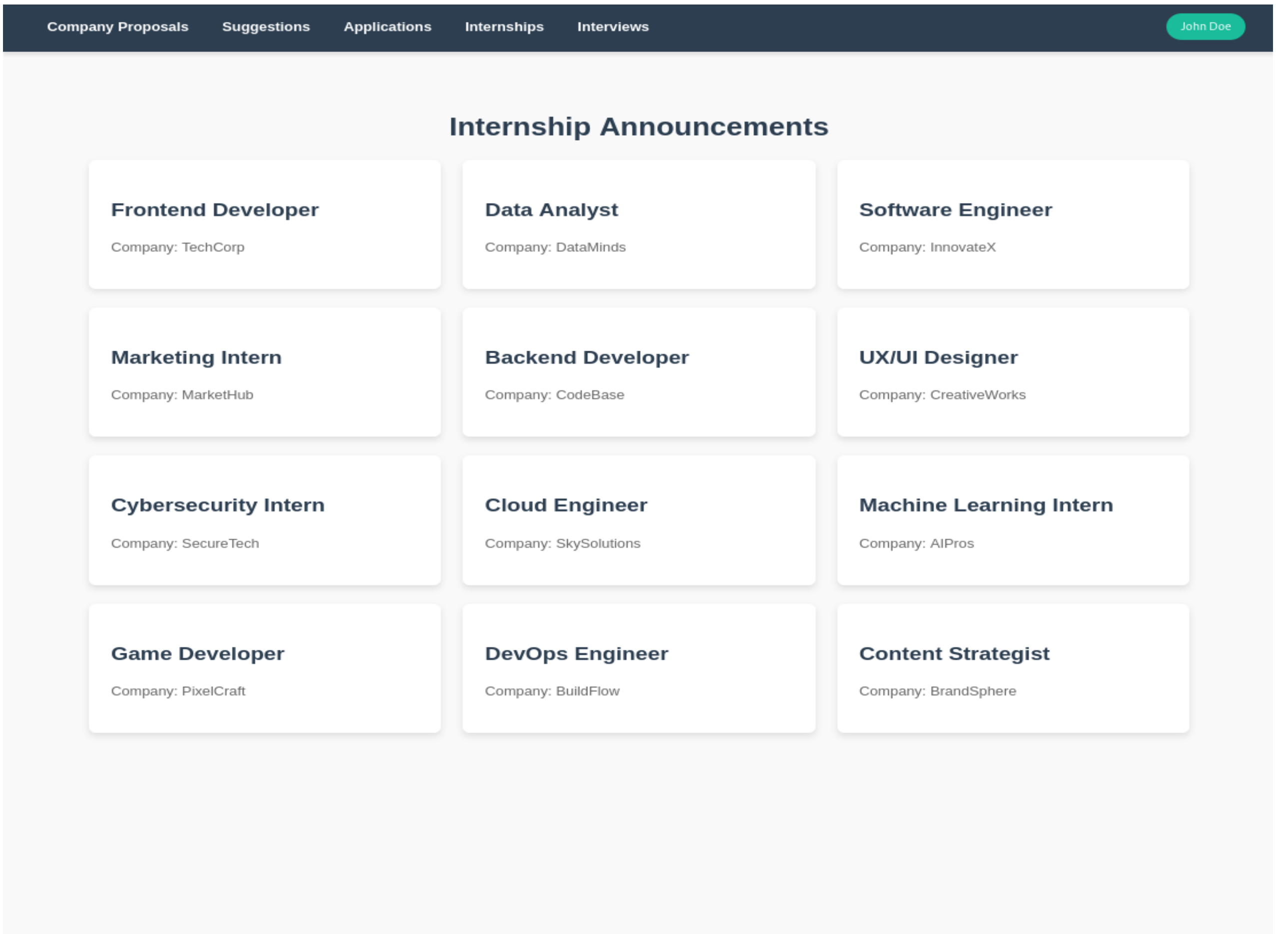
Chapter 3

Specific Requirements

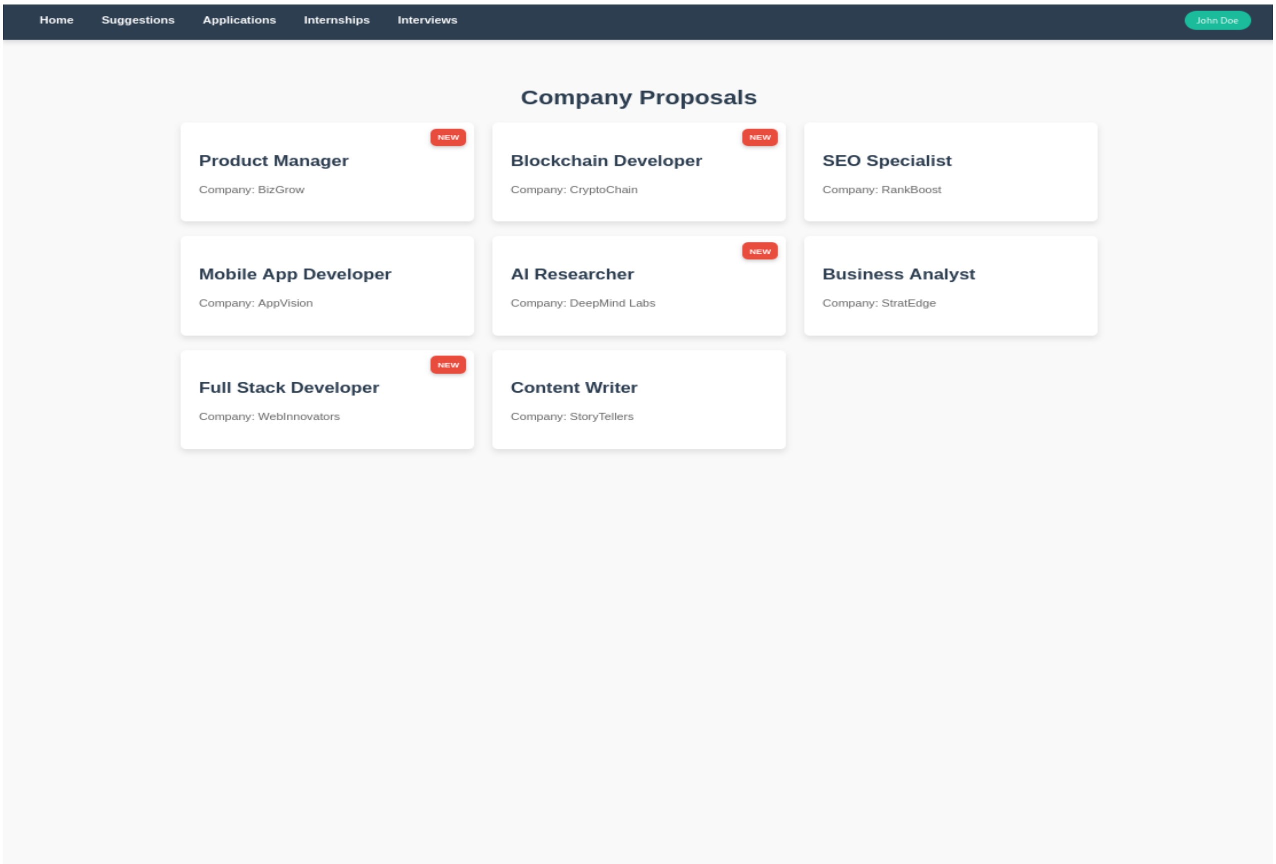
3.1 External Interface Requirements

3.1.1 User Interfaces

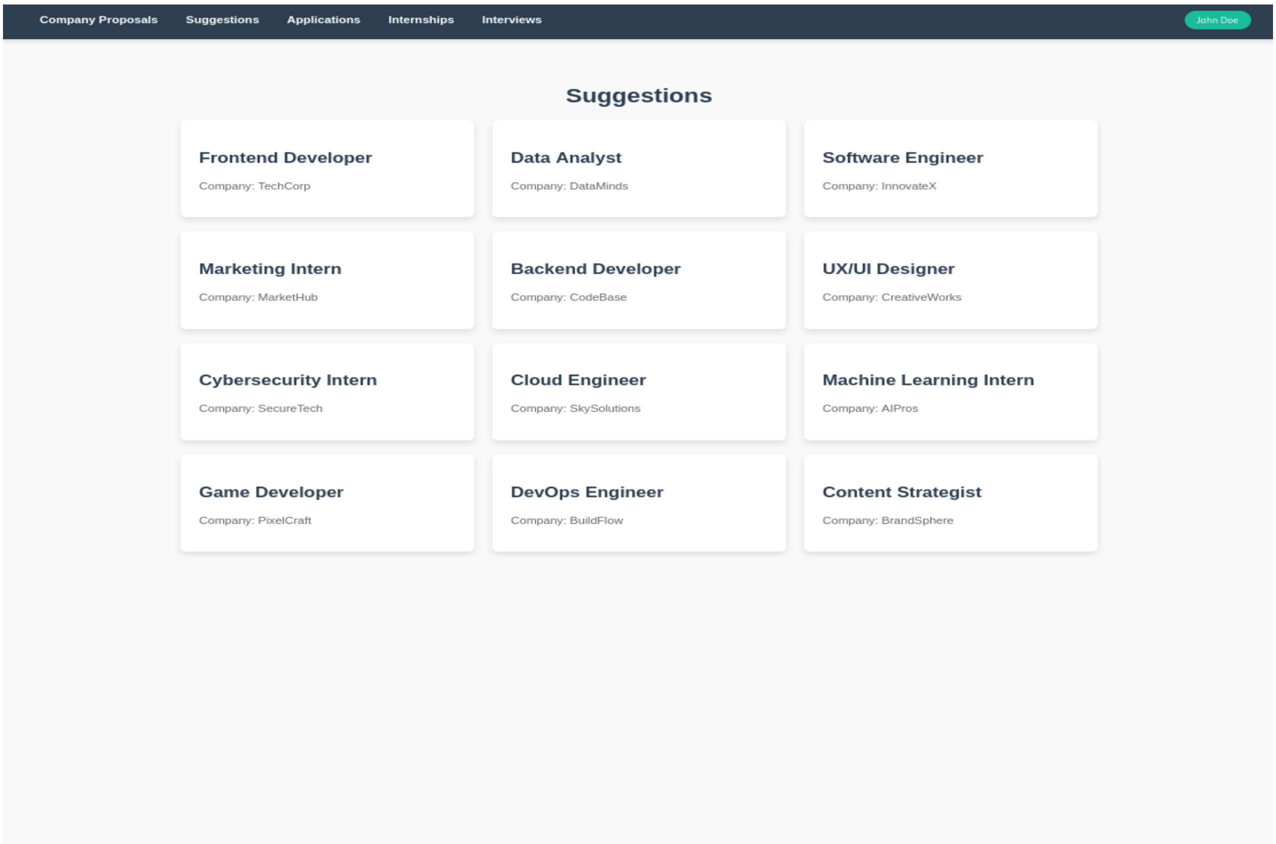
Student's View



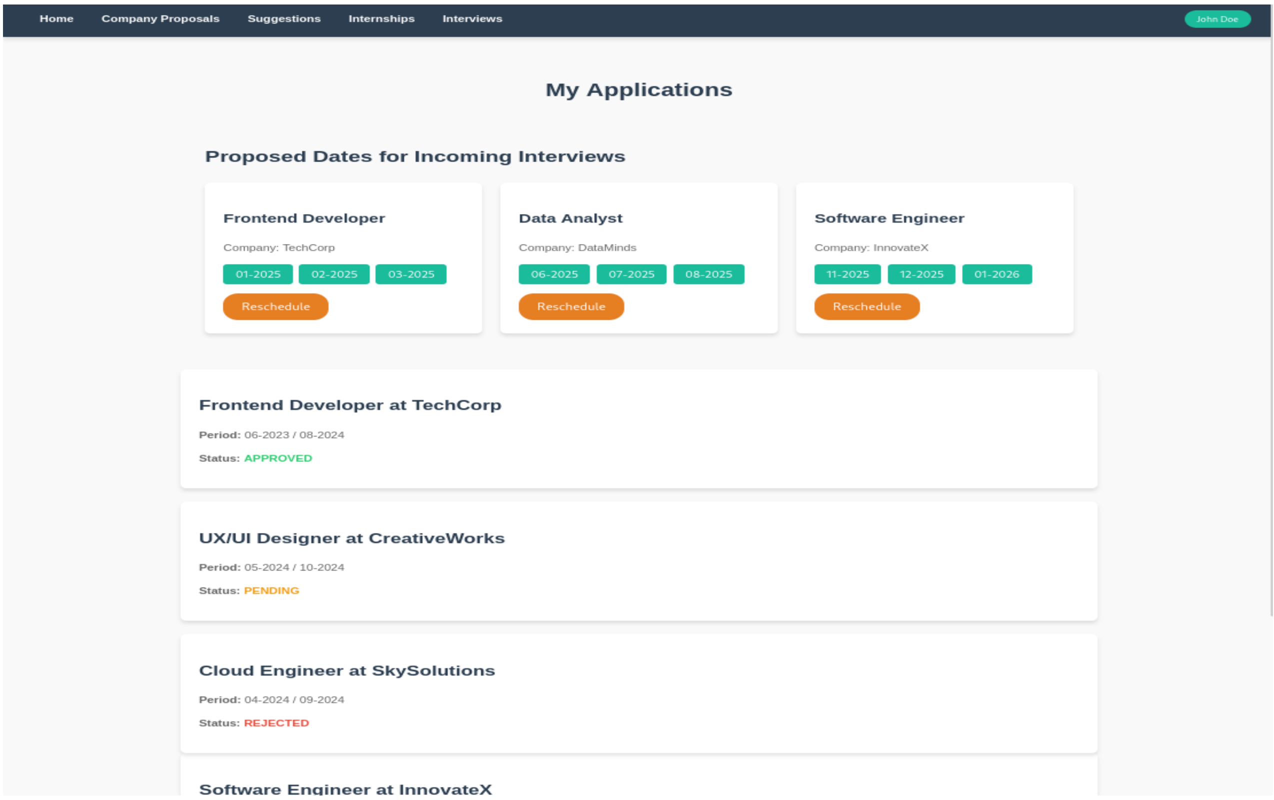
Homepage



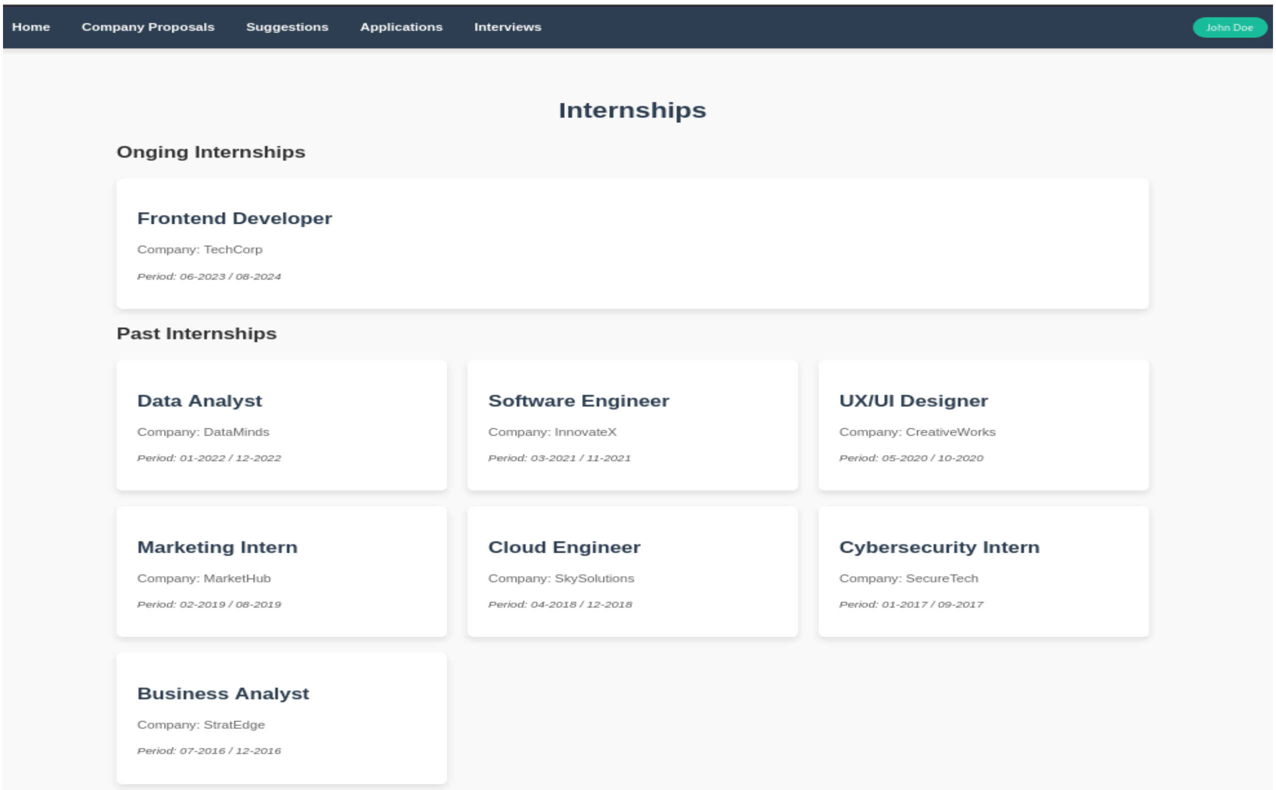
Company Proposals



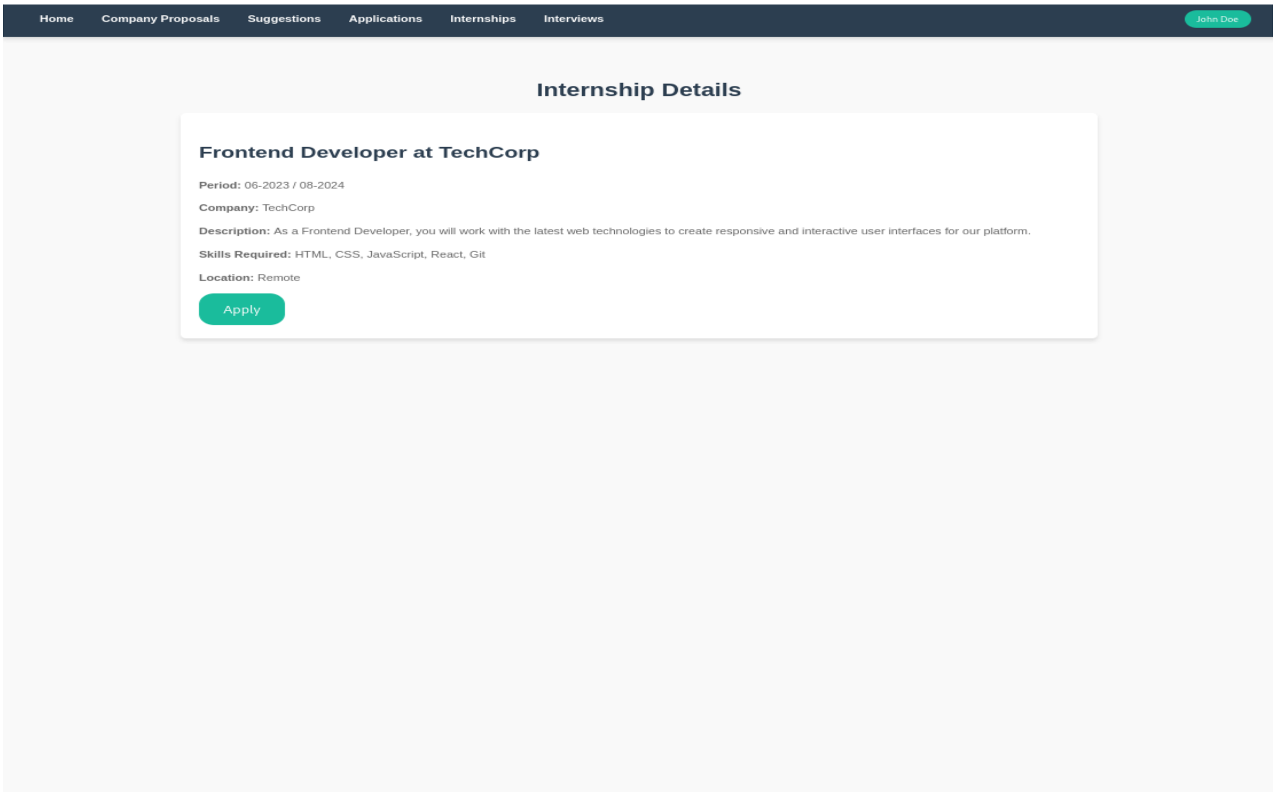
Suggestions



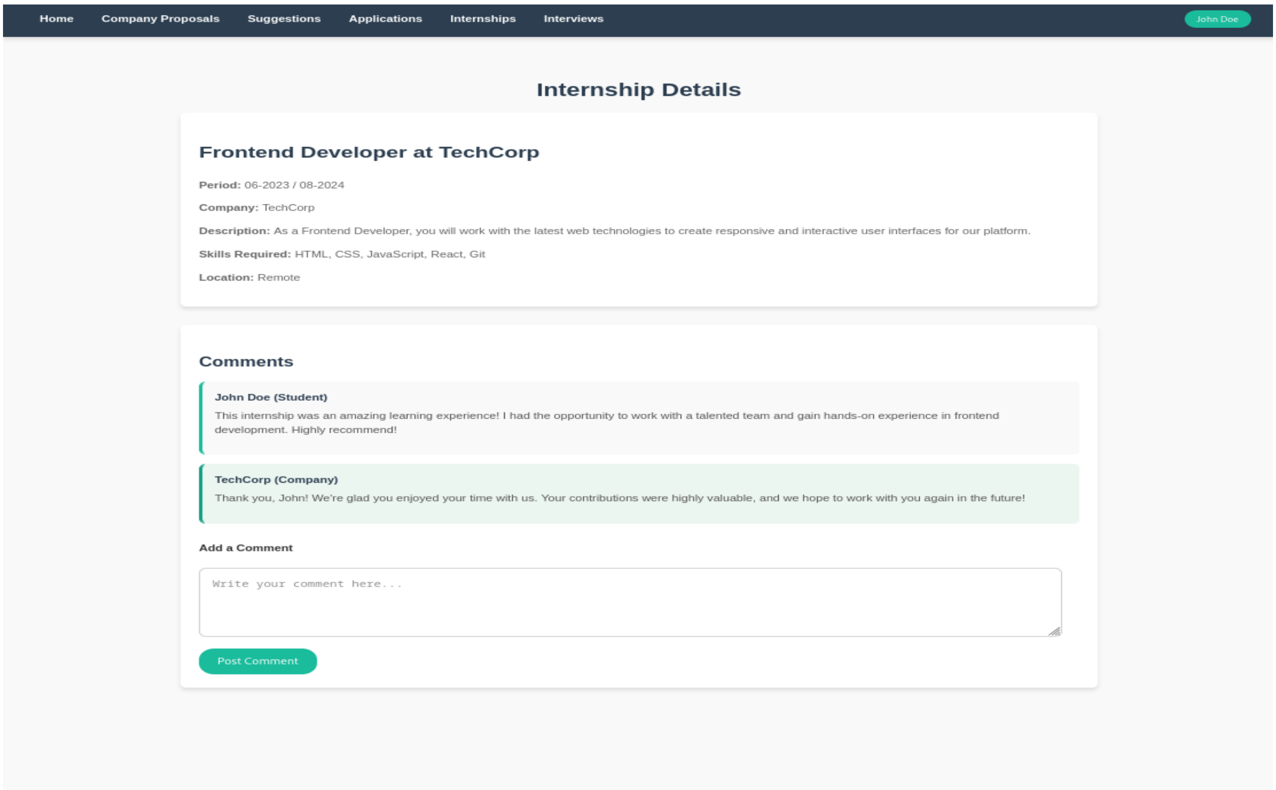
Applications



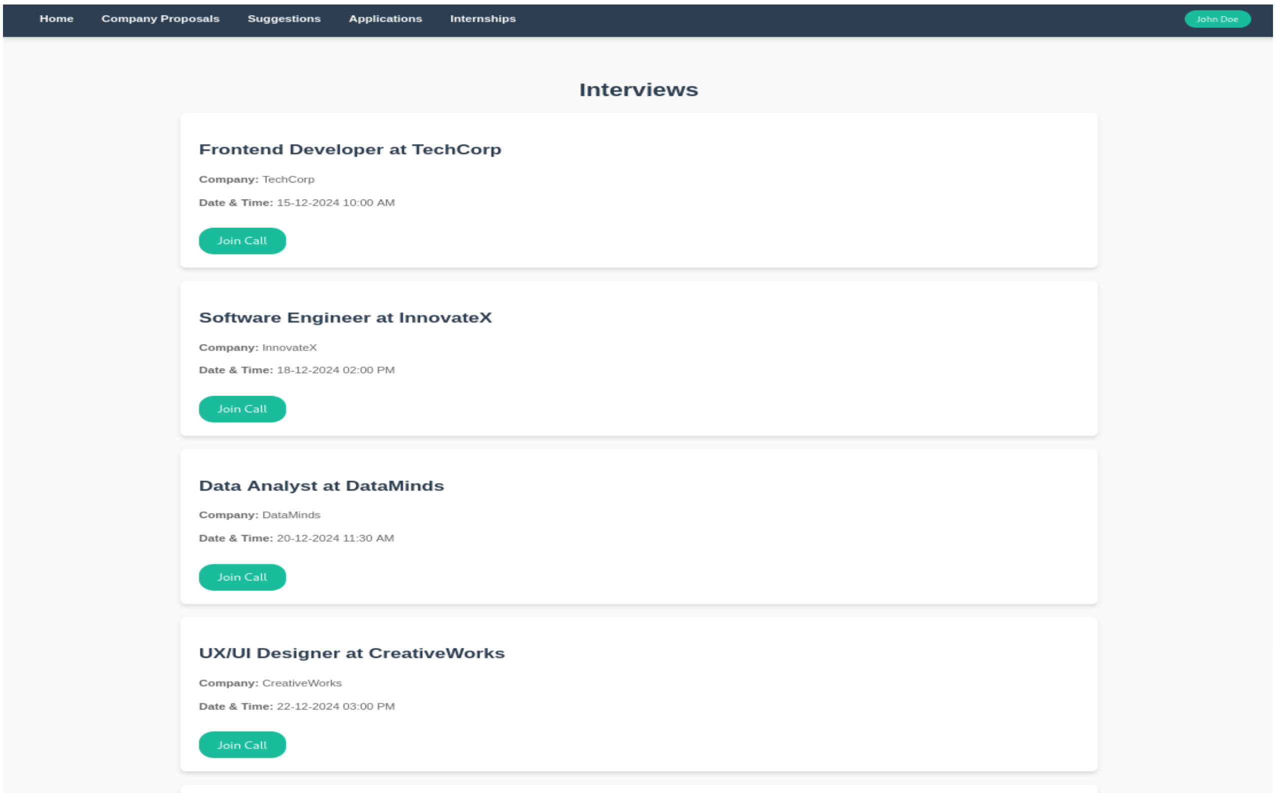
Internships



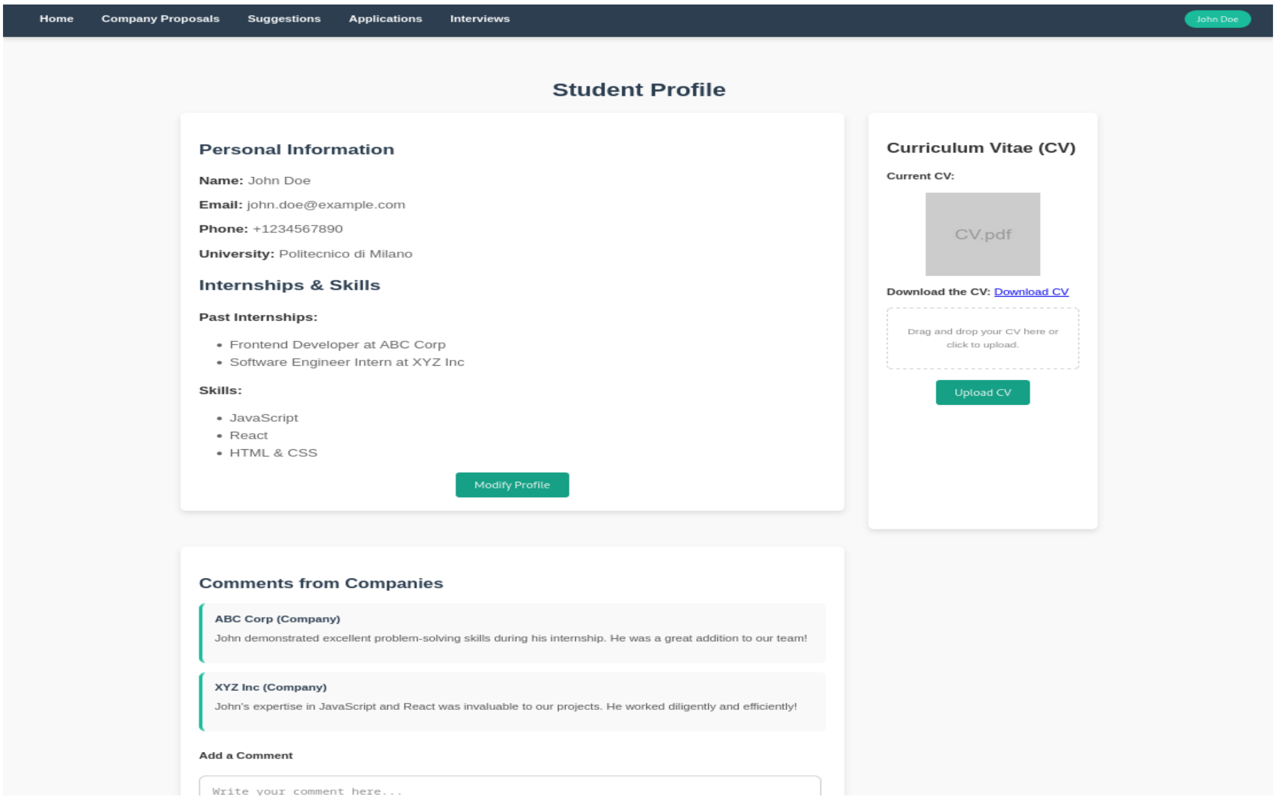
Available internship details



Past internship details

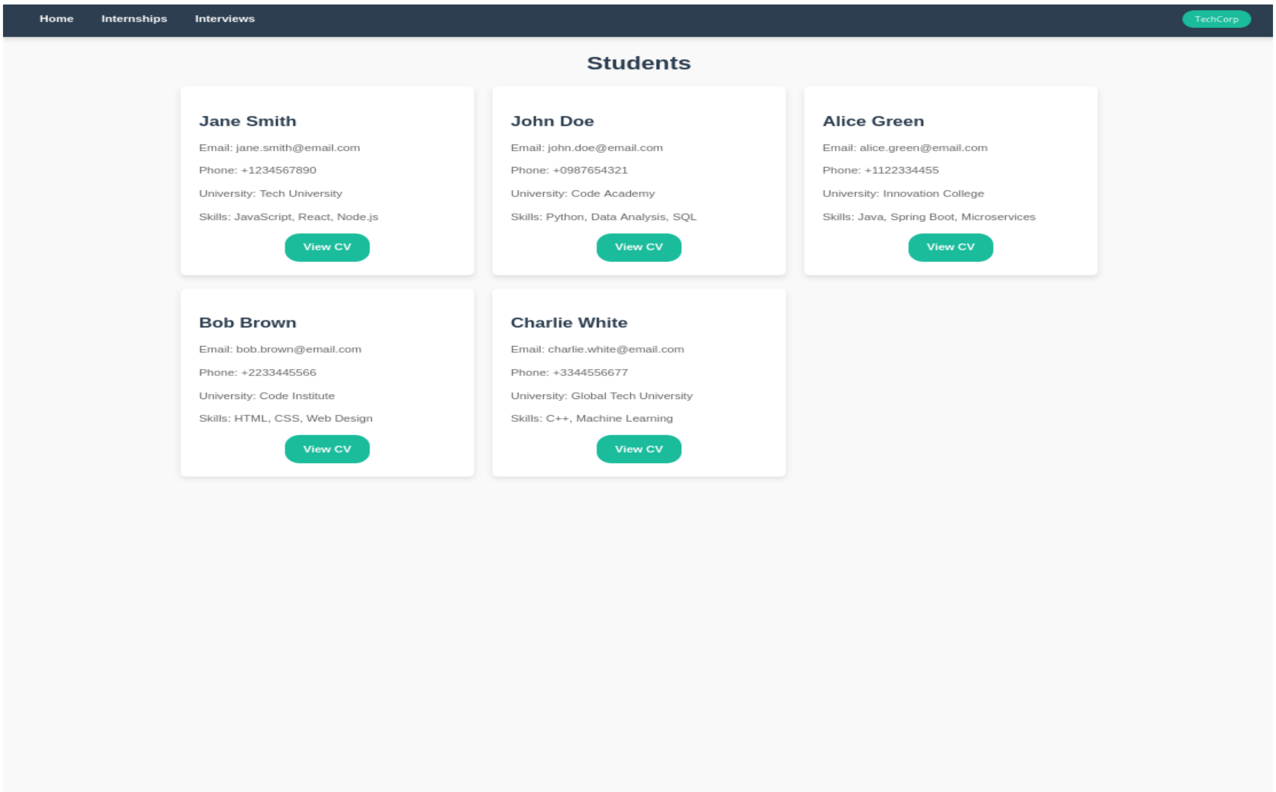


Interviews

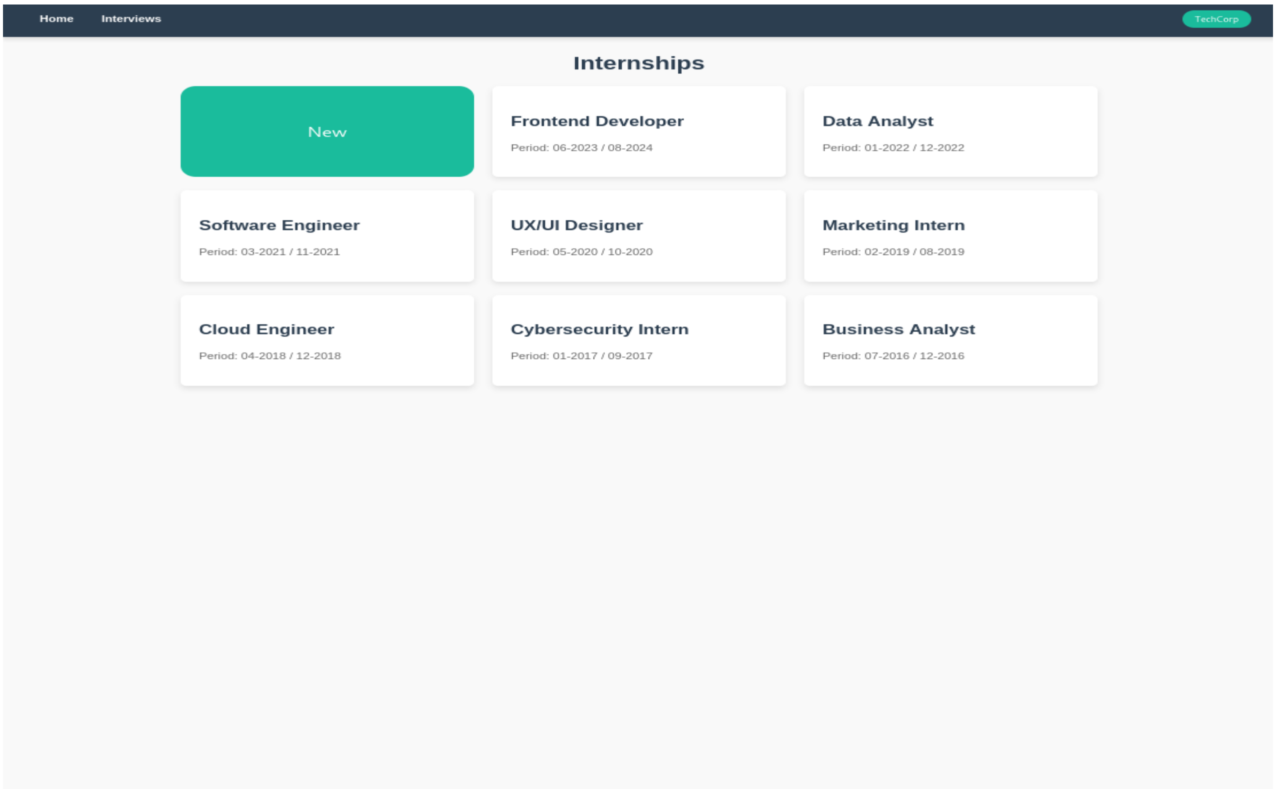


Profile

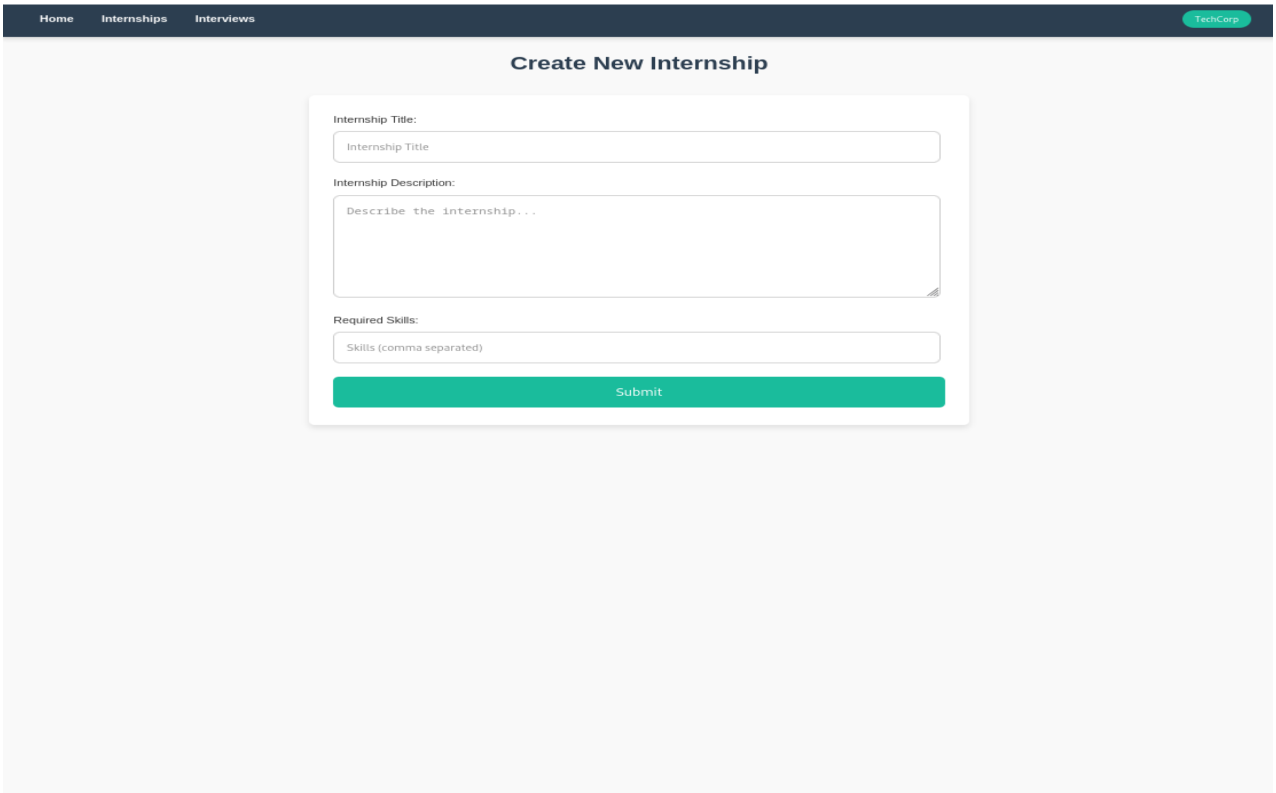
Company's View



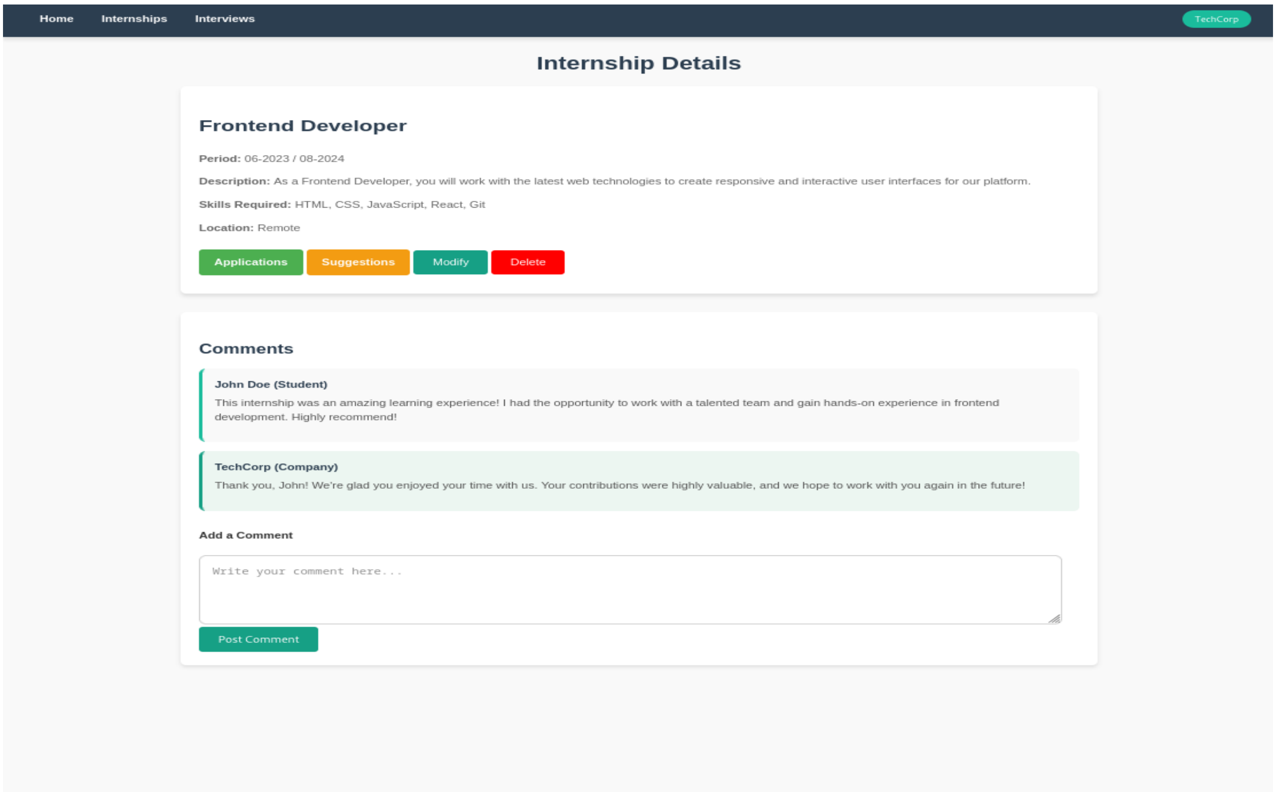
Homepage



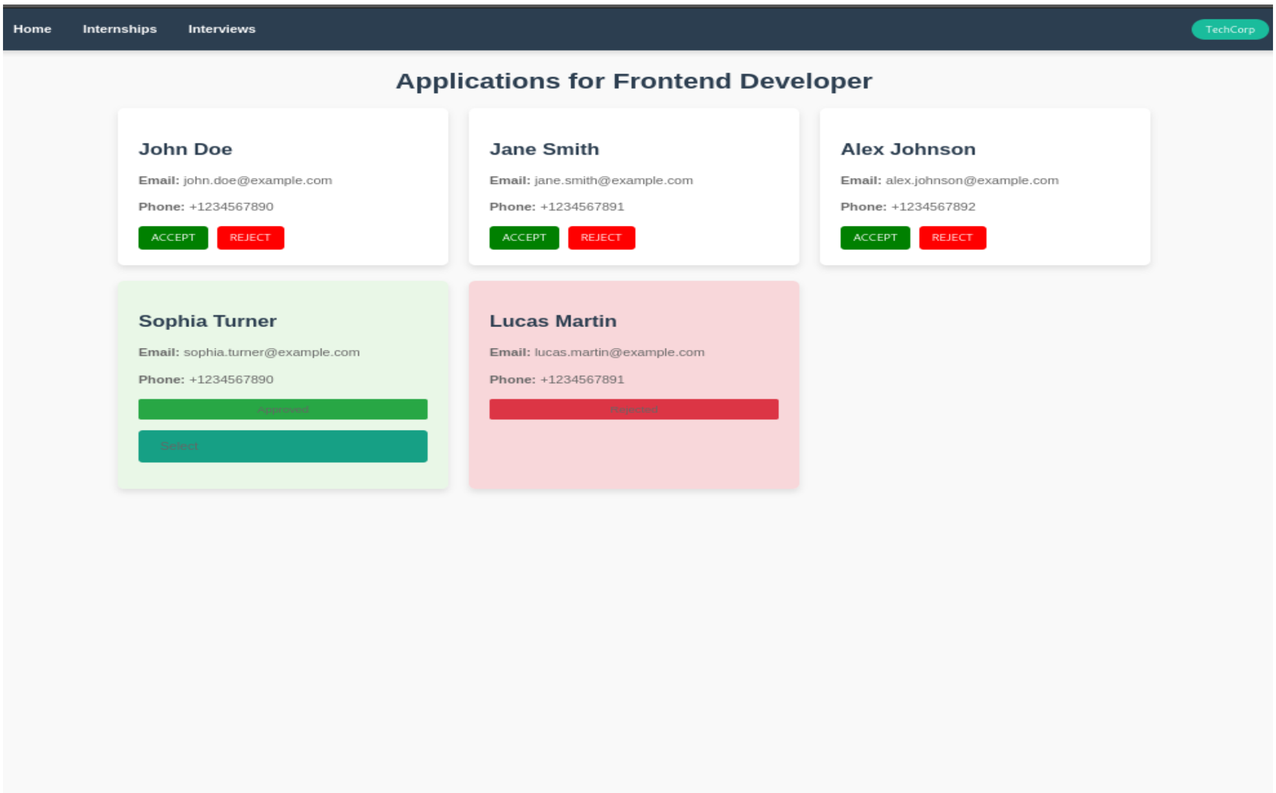
Internships



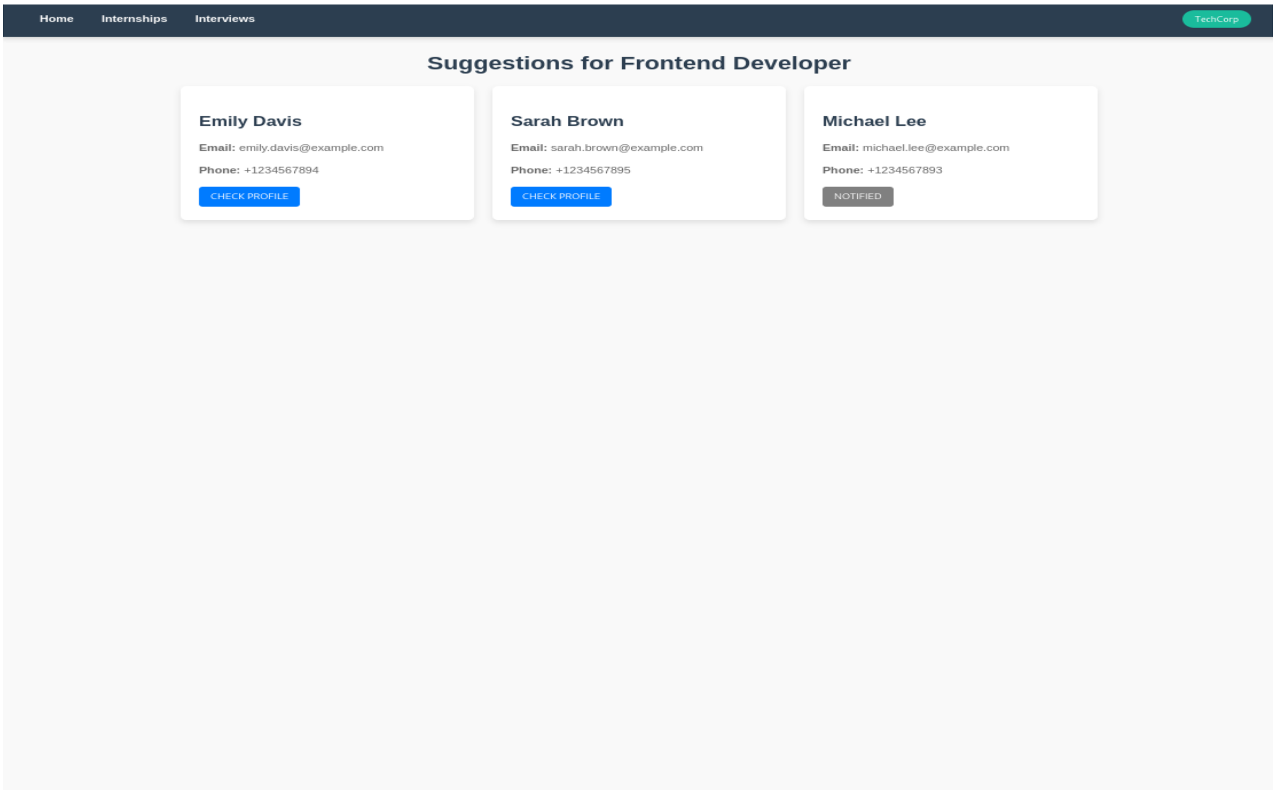
New internship form



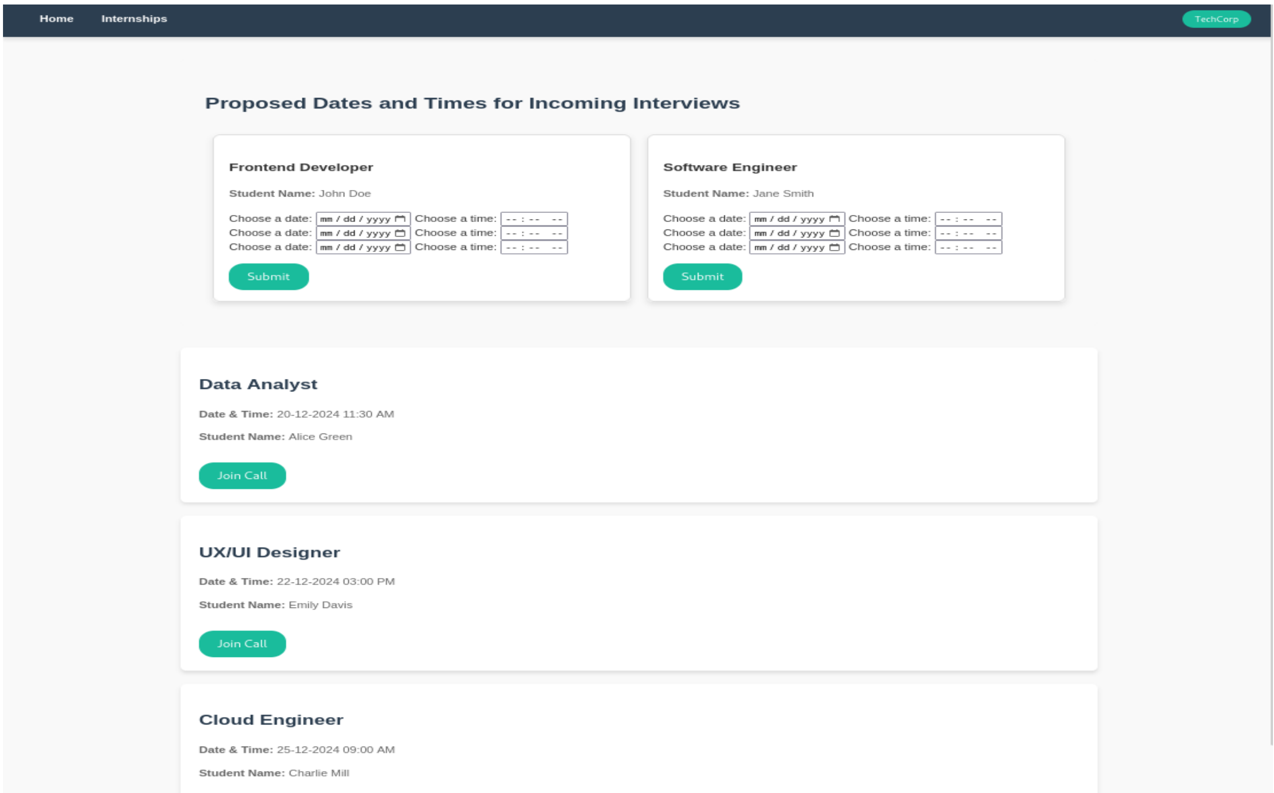
Internship details



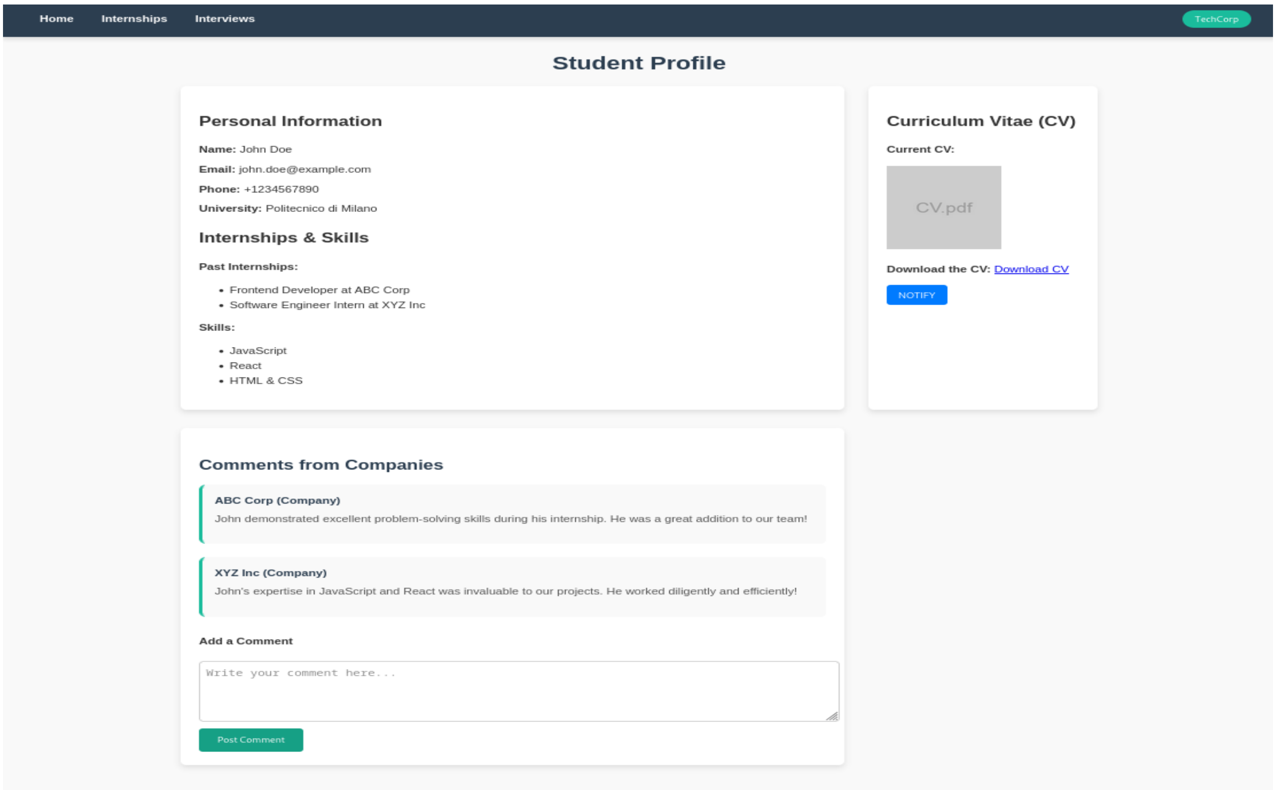
Internship applications



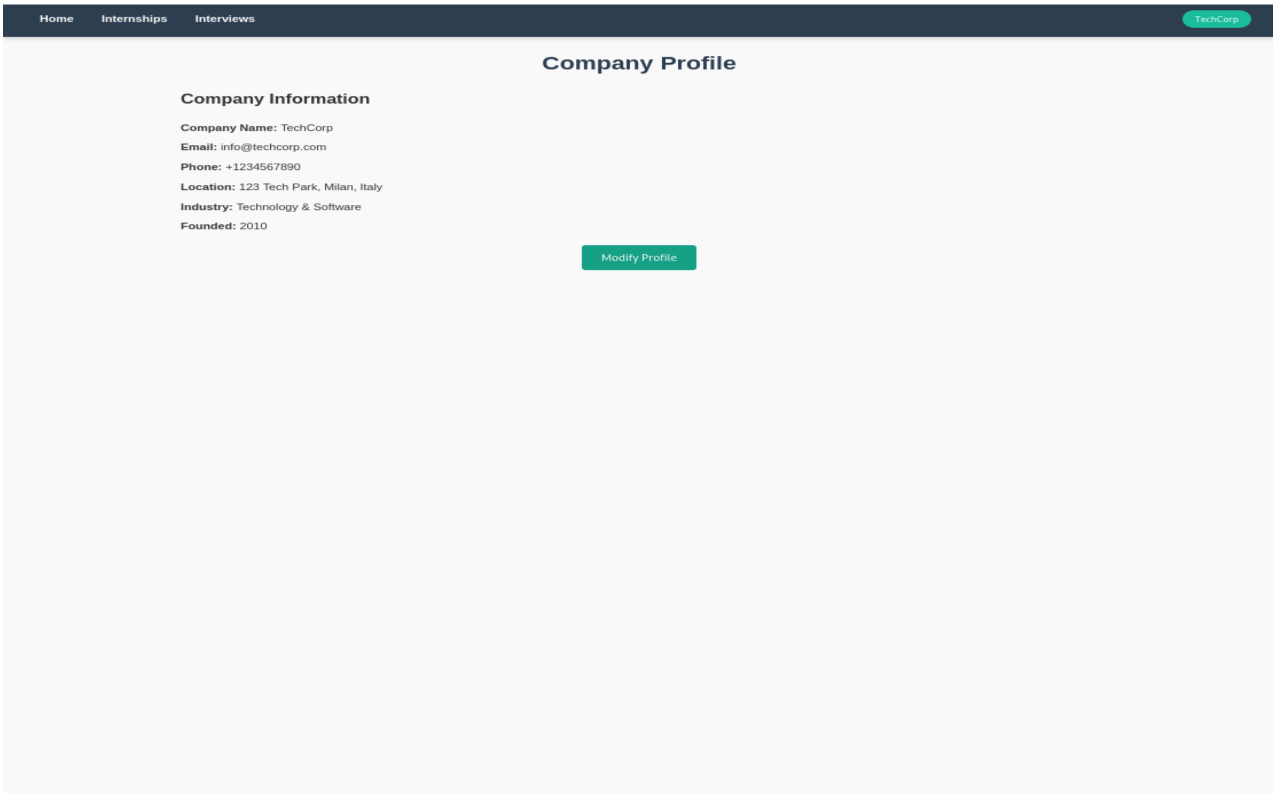
Internship suggestions



Interviews



Student profile



Profile

3.1.2 Hardware Interfaces

S&C is a web app, consequently, it requires a device with a web browser installed to be accessible.

3.1.3 Software Interfaces

Here's a list of the software interfaces needed for the system to properly work:

- Recommendation Engine API: analyzes students' CVs and internship announcements to provide suggestions. It matches students with relevant companies based on their skills and experiences, while also recommending internships to students that align with their qualifications and career goals
- Meeting Application API: offers a web interface for conducting online interviews

3.1.4 Communication Interfaces

The user uses an internet connection to access the platform. The platform must be HTTPS compliant in order to work on the web properly and to be safe.

3.2 Functional Requirements

Sign up and log in

[R1] The System shall allow new users to sign up by providing their personal information

[R2] The System shall allow registered users to log in by providing a username and a password

Student CV management

[R3] The System shall allow Students to upload a CV

Internship announcement posting

[R4] The System shall allow Companies to create internship announcement posts containing all the needed details

[R5] The System shall allow Companies to modify existing internship announcement posts updating the details

[R6] The System shall allow Companies to delete existing internship announcement posts

Internship application

[R7] The System shall allow Students to view the details of any internship

[R8] The System shall allow Students to browse the suggested internships

[R9] The System shall allow Students to apply for an internship

[R10] The System shall allow Students to accept an internship proposal from a Company

[R11] The System shall allow Students to check the status of the applications they've submitted

Student and Company recommendation

[R12] The System shall use the Recommendation Engine to analyze student profiles to match them with internships proposed by companies

[R13] The System shall suggest to Companies Students who match their requirements for an internship proposal

[R14] The System shall suggest to Students Companies that offer internships aligned with their skills and experiences

Company reach out

[R15] The system shall allow Companies to review the suggested students to select the ones they wish to contact for an internship proposal

[R16] The system shall notify Students selected by Companies for an internship

Interview hosting

[R17] The System shall allow Companies to propose three possible dates to Students for an interview

[R18] The System shall allow Students to accept one of the proposed dates for an interview

[R19] The System shall allow Students to ask Companies to provide new possible dates for an interview

[R20] The System shall allow the hosting of online interviews between Companies and Students via a third party app

Student admission

[R21] The System shall allow companies to select students to admit to internships they interviewed for

[R22] The System shall allow students to check for their admission to internships they interviewed for

Student feedback submission

[R23] The System shall ask Students to provide feedback about an internship by filling out a form

[R24] The System shall allow Companies to publicly comment on the feedback they received

[R25] The System shall allow Students to see feedback about Companies from past internships they offered

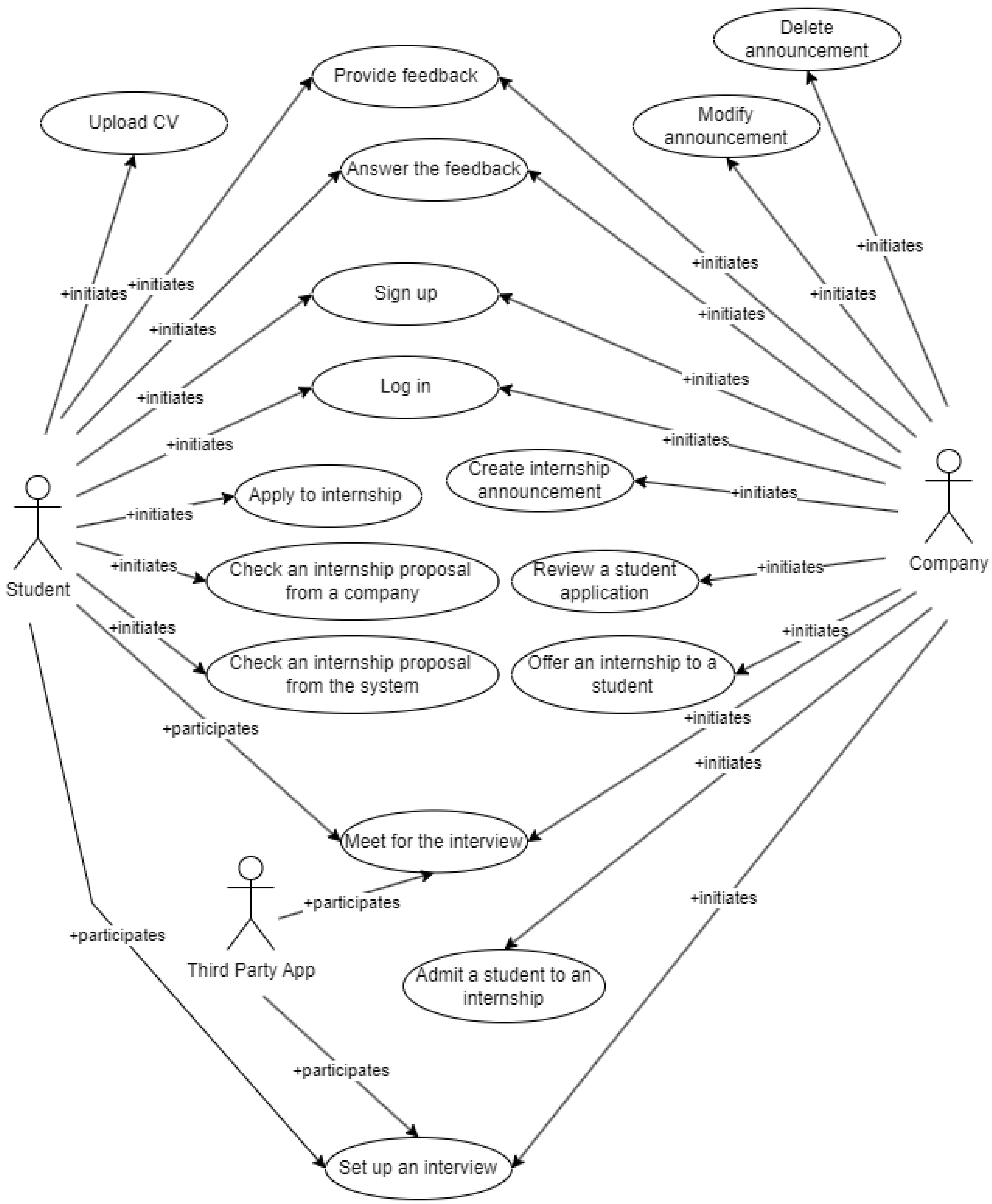
Company feedback submission

[R26] The System shall ask Companies to provide feedback about the Student's performance by filling out a form

[R27] The System shall allow Students to publicly comment on the feedback they received

[R28] The System shall allow Companies to see feedback about Students from past internships they partook in

3.2.1 Use Cases Diagram



3.2.2 Use Cases

Attribute	**Description**
Name	User signs up
Actors	• User
Entry Condition	• The user has opened the S&C platform's landing page
Event Flow	1. The user presses the "Sign up" button 2. The user selects its role by pressing either the "Company" or the "Student" button 3. The user fills out a registration form 4. The user submits the form
Exit Condition	• The user is successfully registered on the platform with his new credentials and is redirected to the homepage
Exceptions	5. A user with these credentials already exists. The system sends a warning and does not allow the signup process to continue, redirecting the user to the "Log in" page

[UC1] User signs up

Attribute	Description
Name	User logs in
Actors	• User
Entry Condition	• The user has opened the S&C platform's landing page • The student is already signed up to the platform
Event Flow	1. The user presses the "Log in" button 2. The user fills out the login form with his credentials 3. The user submits the form
Exit Condition	• The user has successfully logged in to the platform
Exceptions	4a. A user with these credentials does not exist. The system sends a warning and the login process fails. The user is redirected to the "Sign up" page 4b. A user with these credentials exists but the password is incorrect. The system sends a warning to the user, and after a certain number of incorrect tries the login process is halted by the system

[UC2] User logs in

Attribute	Description
Name	Student uploads CV
Actors	• Student
Entry Condition	• The student has logged in to the S&C platform
Event Flow	1. The student presses the "Profile" button 2. The student uploads the CV via the file system 3. The student presses the "Upload CV" button
Exit Condition	• The user has successfully uploaded the CV
Exceptions	4a. The file is not a PDF. The system sends a warning to the user

[UC3] Student uploads CV

Attribute	Description
Name	Student applies to an internship he found
Actors	• Student
Entry Condition	• The student has logged in to the S&C platform's
Event Flow	1. The student browsing the available internships on the homepage finds one that suits him 2. The student presses on the internship, and a view with its details opens 3. The student presses the "Apply" button
Exit Condition	• The student has successfully applied to the internship, his application status is PENDING
Exceptions	

[UC4] Student applies to an internship he found

Attribute	**Description**
Name	Student checks an internship proposed by a company
Actors	• Student
Entry Condition	• The student has logged in to the S&C platform • A company has selected the student and proposed an internship through the platform
Event Flow	1. The student presses the "Company proposals" button 2. The student presses the internship that is tagged as "new" in the list of internships suggested by companies, and a view with the internship's details opens 3. The student decides whether or not to accept the internship, by clicking on the "Apply" button or not
Exit Condition	• The student has successfully applied to the internship, their application status is APPROVED
Exceptions	

[UC5] Student checks an internship proposed by a company

Attribute	**Description**
Name	Student checks an internship proposed by the system
Actors	• Student
Entry Condition	• The student has logged in to the S&C platform
Event Flow	1. The student presses the "Suggestions" button 2. The student presses on one of the available internships, and a view with the internship's details opens 3. The student decides whether or not to accept the internship, by pressing either the "Apply" button or not
Exit Condition	• The student has successfully applied to the internship, his application status is PENDING
Exceptions	

[UC6] Student checks an internship proposed by the system

Attribute	Description
Name	Company creates an internship announcement
Actors	• Company
Entry Condition	• The company has logged in to the S&C platform
Event Flow	1. The company presses the "Internships" button 2. The company presses the "New" button 3. The company fills the form with the internship's details 4. The company presses the "Submit" button
Exit Condition	• The company has successfully created an internship announcement
Exceptions	5.a The information provided by the company is invalid. The system sends a warning

[UC7] Company creates an internship announcement

Attribute	Description
Name	Company modifies an internship announcement
Actors	• Company
Entry Condition	• The company has logged in to the S&C platform • The company has previously posted the internship announcement
Event Flow	1. The company presses the "Internships" button 2. The company selects the existing internship it whishes to modify and presses on its "Modify" button 3. The company modifies the already filled in announcement 4. The company presses the "Submit" button
Exit Condition	• The company has successfully modified an existing internship announcement
Exceptions	5.a The information provided by the company is invalid. The system sends a warning

[UC8] Company modifies an internship announcement

Attribute	Description
Name	Company deletes an internship announcement
Actors	• Company
Entry Condition	• The company has logged in to the S&C platform • The company has previously posted the internship announcement
Event Flow	1. The company presses the "Internships" button 2. The company selects the existing internship it whishes to delete and presses on its "Delete" button 3. The company confirms its choice by pressing the "Confirm" button that pops up
Exit Condition	• The company has successfully deleted an internship announcement
Exceptions	

[UC9] Company deletes an internship announcement

Attribute	Description
Name	Company reviews a student’s application
Actors	• Company
Entry Condition	• The company has logged in to the S&C platform
Event Flow	1. The company presses the "Internships" button 2. The company selects the internship whose applications it wants to check by pressing it 3. The company presses the "Applications" button 4. The company selects the student whose application it wants to check by pressing it 5. The company decides whether or not to accept the application by pressing either the "Accept" or the "Reject" buttons
Exit Condition	• The company has successfully accepted a student’s application • The company has successfully rejected a student’s application
Exceptions	

[UC10] Company reviews a student’s application

Attribute	Description
Name	Company offers an internship to a student
Actors	• Company
Entry Condition	• The company has logged in to the S&C platform
Event Flow	1. The company presses the "Internships" button 2. The company presses the "Suggestions" button 3. The company browses the suggested students and finds one that suits it 4. The company presses on the student and a view with the student's details opens 5. The company presses the "Notify" button
Exit Condition	• The company has successfully offered the internship to the student
Exceptions	

[UC11] Company offers an internship to a student

Attribute	Description
Name	Student and company set up the interview
Actors	• Student • Company • Third party meeting app
Entry Condition	• The company has posted an internship announcement on the platform • The student has been accepted by the company for the interview • The company has logged in to the S&C platform • The student has logged in to the S&C platform
Event Flow	1. The company goes to the "Interviews" section 2. The company selects three possible dates from a calendar and presses the "Submit" button associated with the student's name 3. The student enters the "Applications" section and selects an internship to which they have been admitted to 4. The student chooses one of the dates by pressing one of the three buttons relative to the three dates 5. The third party meeting app schedules the meeting and sends the link for it to both parties
Exit Condition	• The meeting has been scheduled and both the student and the company have been sent the link
Exceptions	4a. The student is not available on any of the dates and thus requests a new batch of dates by pressing on the "Reschedule" button. The company receives a message telling it to propose three new dates

[UC12] Student and company set up the interview

Attribute	Description
Name	Student and company meet for the interview
Actors	• Student • Company • Third party meeting app
Entry Condition	• The company and the student have agreed on a date for the interview • The student has logged in to the S&C platform • The company has logged in to the S&C platform
Event Flow	1. The company presses the "Interviews" button 2. The company presses on the third party meeting app link associated with the student's name 3. The student presses the "Interviews" button 4. The student presses on the third party meeting app link associated with the company's name
Exit Condition	• The student and the company have successfully started the interview via the third party meeting app
Exceptions	

[UC13] Student and company meet for the interview

Attribute	**Description**
Name	Company admits a student to an internship
Actors	• Company
Entry Condition	• The company has logged in to the S&C platform • The company interviewed the student it wants to admit
Event Flow	1. The company presses the "Internships" button 2. The company selects the internship they want to admit a student to 3. The company presses the "Applications" button 4. The company presses the "Select" button of the student it wants to admit to the internship
Exit Condition	• The student has successfully been admitted to the internship
Exceptions	

[UC14] Company admits a student to an internship

Attribute	Description
Name	Student provides feedback on an internship
Actors	• Student
Entry Condition	• The student has logged in to the S&C platform • The student took part in the internship they want to provide feedback about
Event Flow	1. The student presses the "Internships" button, and a view with the internships they took part in opens 2. The student selects the internship they want to provide feedback about, a view with the internship's details opens 3. The student writes the feedback in the comments section 4. The student presses the "Submit" button
Exit Condition	• The student has successfully provided feedback about the internship they took part in
Exceptions	• The feedback surpasses the character limits for comments. The system sends a warning

[UC15] Student provides feedback on an internship

Attribute	Description
Name	Company provides feedback on a student
Actors	• Company
Entry Condition	• The company has logged in to the S&C platform • The company offered the internship to the student It wants to provide feedback about taking part in
Event Flow	1. The company presses the "Internships" button, and a view with the internships they created opens 2. The company selects the internship, and a view with the internship's details opens 3. The company selects the student from the list of participants, and a view with the student's details opens 4. The company writes the feedback in the comments section 5. The company presses the "Submit" button
Exit Condition	• The company has successfully provided feedback about a student involved in one of its internships
Exceptions	• The feedback surpasses the character limits for comments. The system sends a warning

[UC16] Company provides feedback on a student

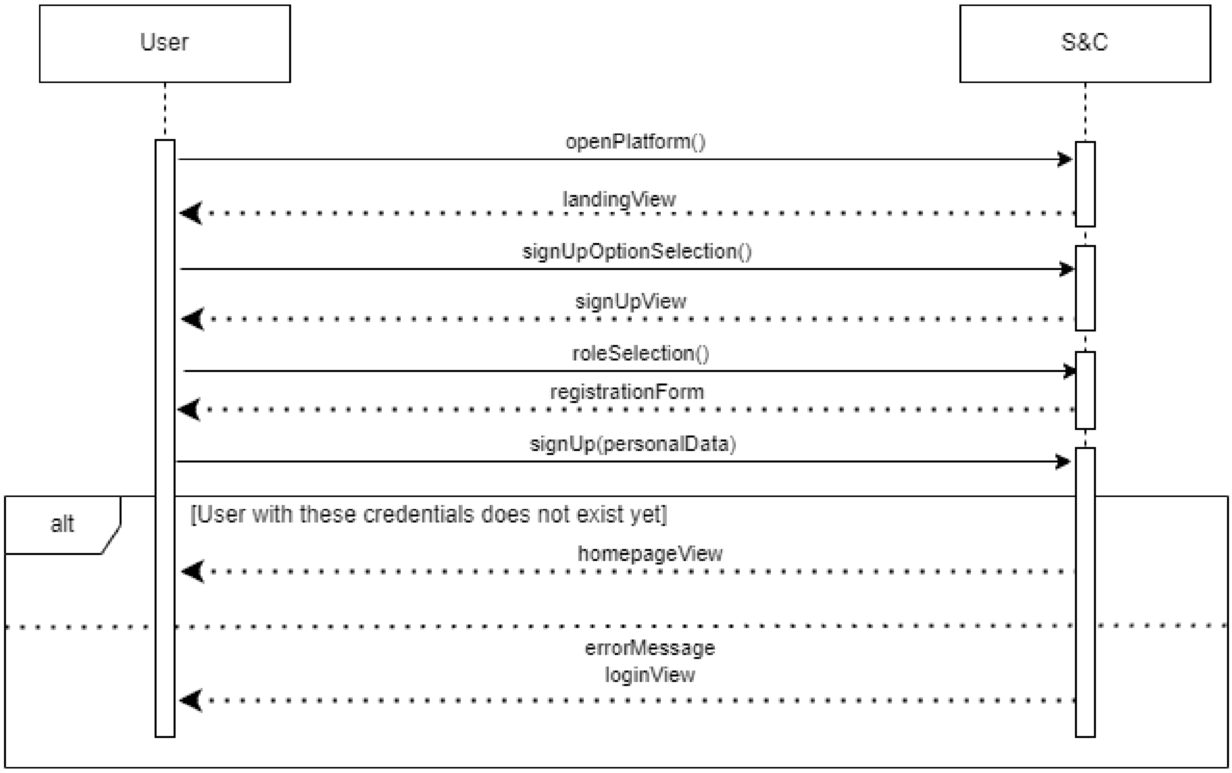
Attribute	Description
Name	Student answers a comment about them
Actors	• Student
Entry Condition	• The student has logged in to the S&C platform • The student received feedback from a company based on an internship they took part in
Event Flow	1. The student presses the "Profile" button, and a view with the student's details opens 2. The student reads the feedback from the company whose internship they partook in 3. The student writes the answer to the feedback in the comments section 4. The student presses the "Submit" button
Exit Condition	• The student has successfully provided an answer to feedback provided by a company whose internship they partook in
Exceptions	• The answer surpasses the character limits for comments. The system sends a warning

[UC17] Student answers a comment about them

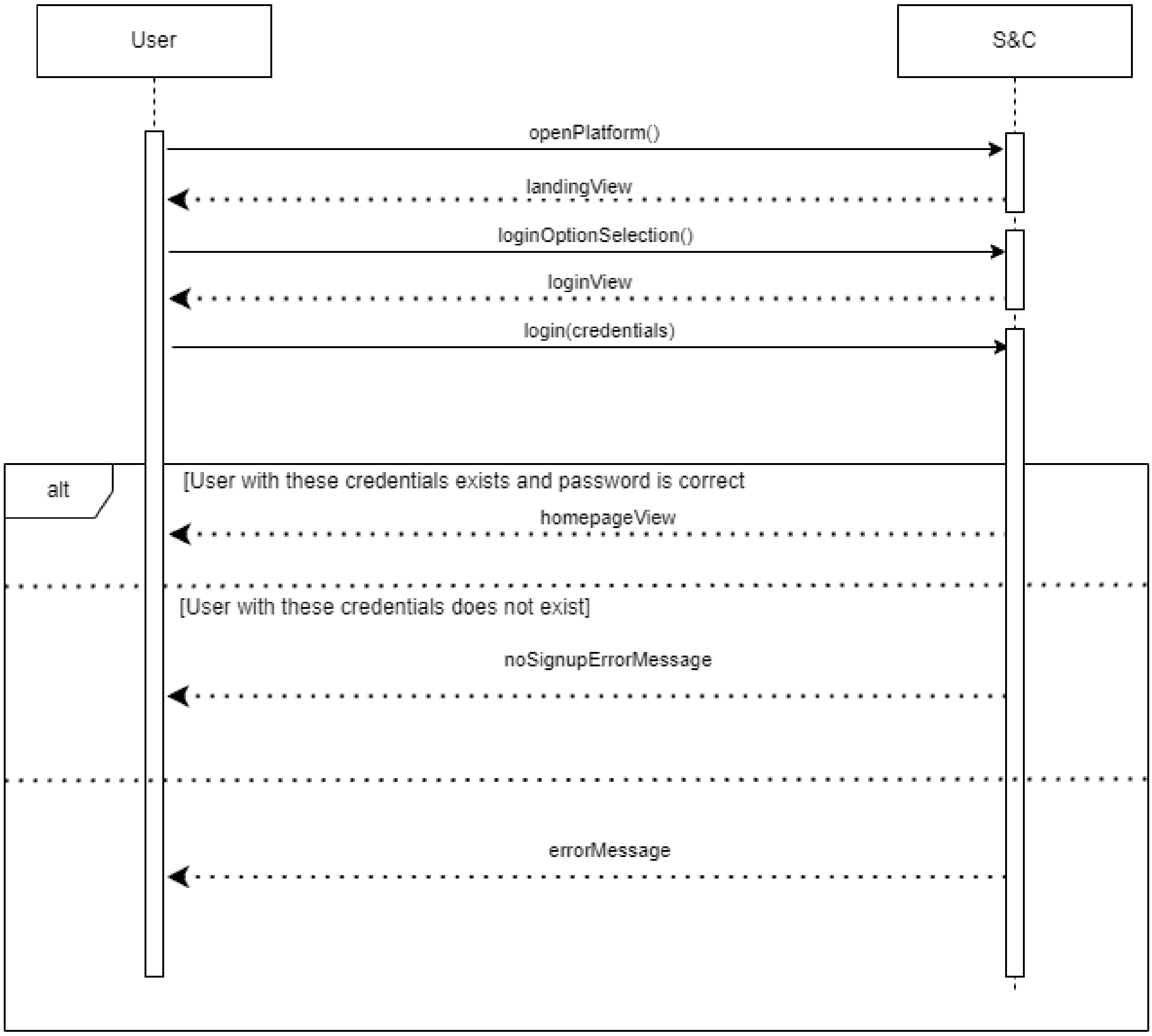
Attribute	Description
Name	Company answers a comment about an internship it offered
Actors	• Company
Entry Condition	• The company has logged in to the S&C platform • The company received feedback from a student about an internship it offered
Event Flow	1. The company presses the "Internships" button, and a view with the internships offered by the company opens 2. The company selects an internship by pressing it, a view with the internship's details opens 3. The company reads the feedback about the internship from a student who took part in it 4. The company writes the answer to the feedback in the comments section 5. The company presses the "Submit" button
Exit Condition	• The company has successfully provided an answer to feedback about an internship it offered provided by a student who took part in it
Exceptions	• The answer surpasses the character limits for comments. The system sends a warning

[UC18] Company answers a comment about an internship if offered

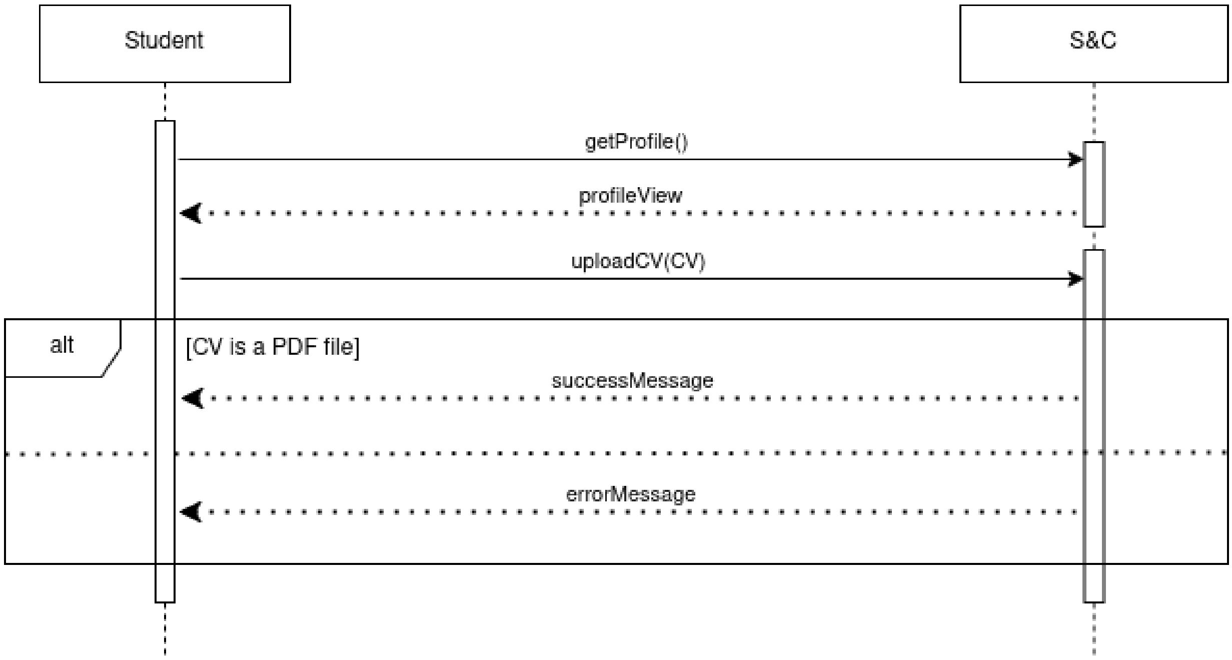
3.2.3 Sequence Diagrams



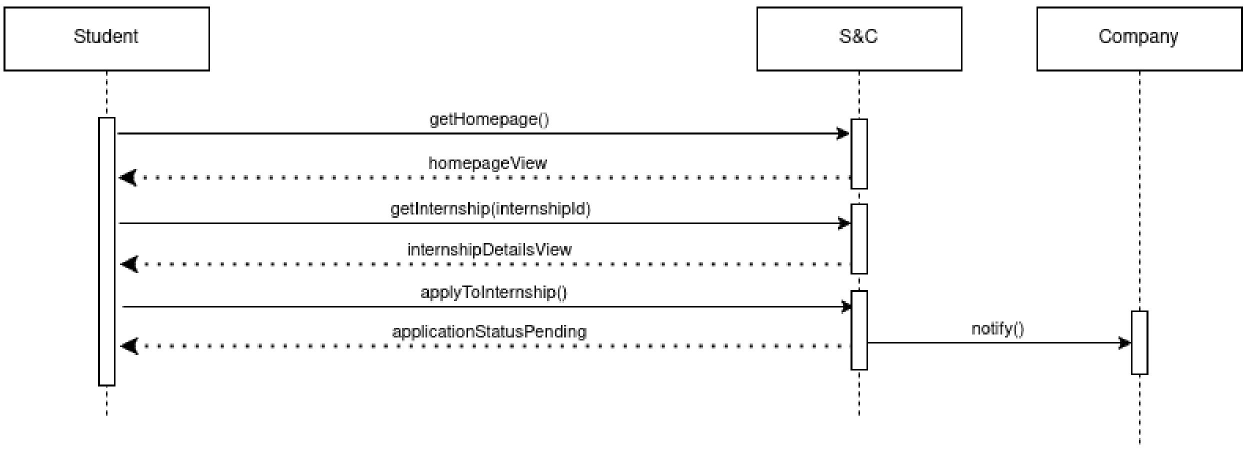
[UC1] User signs up



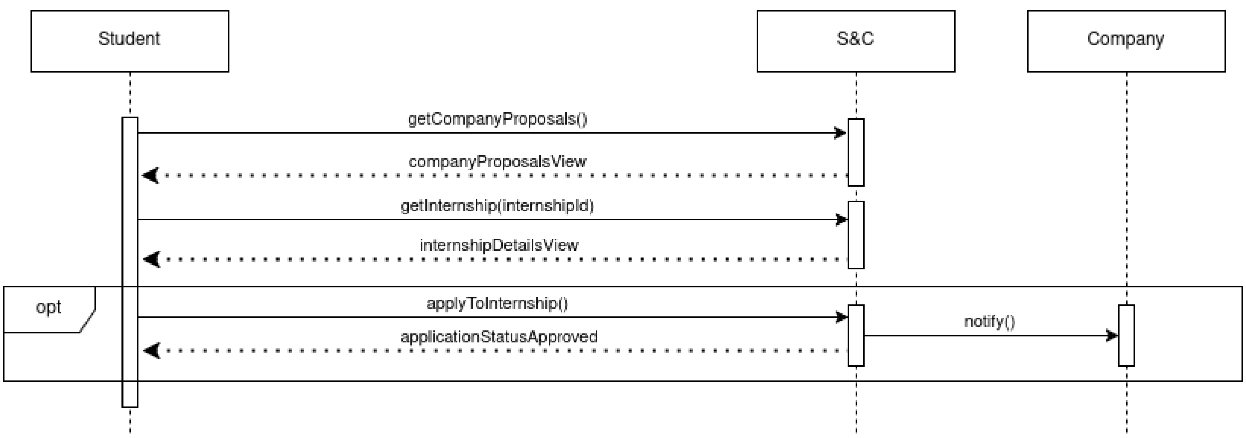
[UC2] User logs in



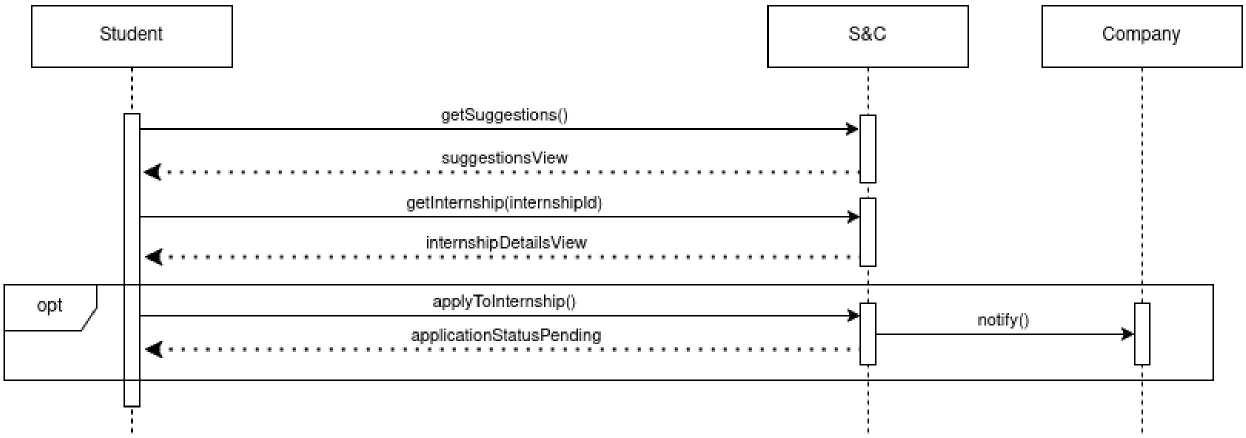
[UC3] Student uploads CV



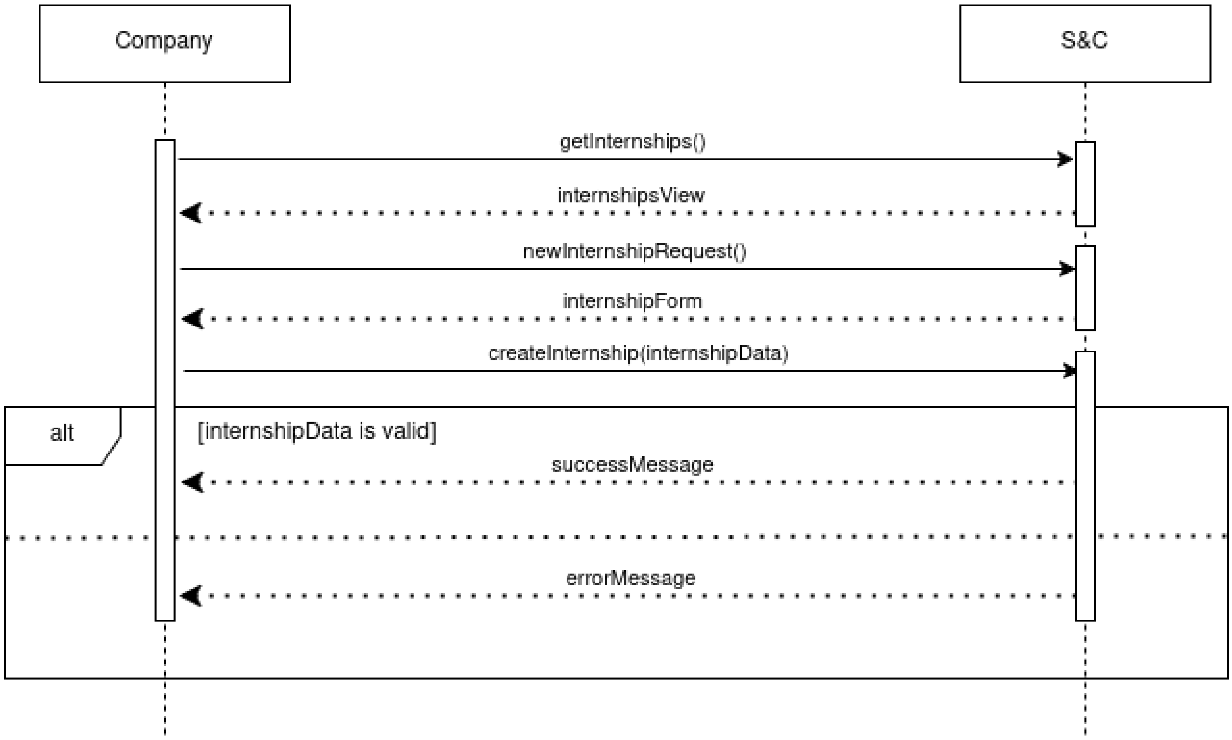
[UC4] Student applies to an internship he found



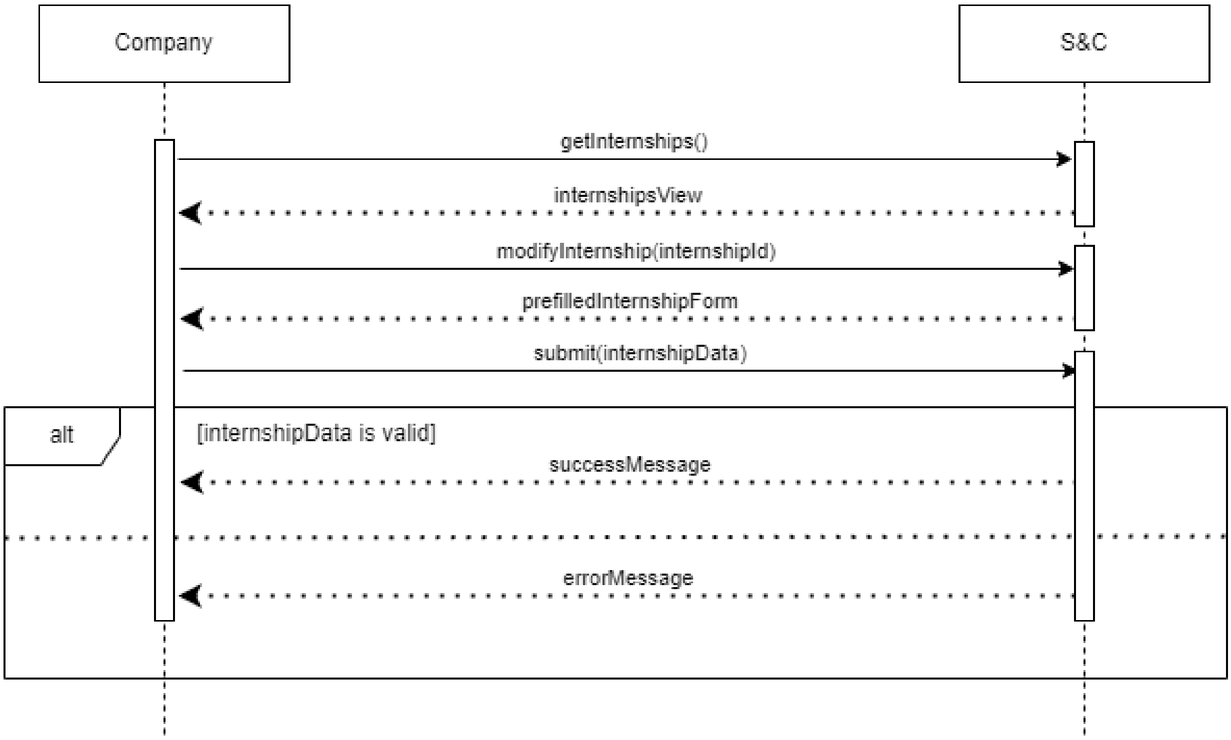
[UC5] Student checks an internship proposed by a company



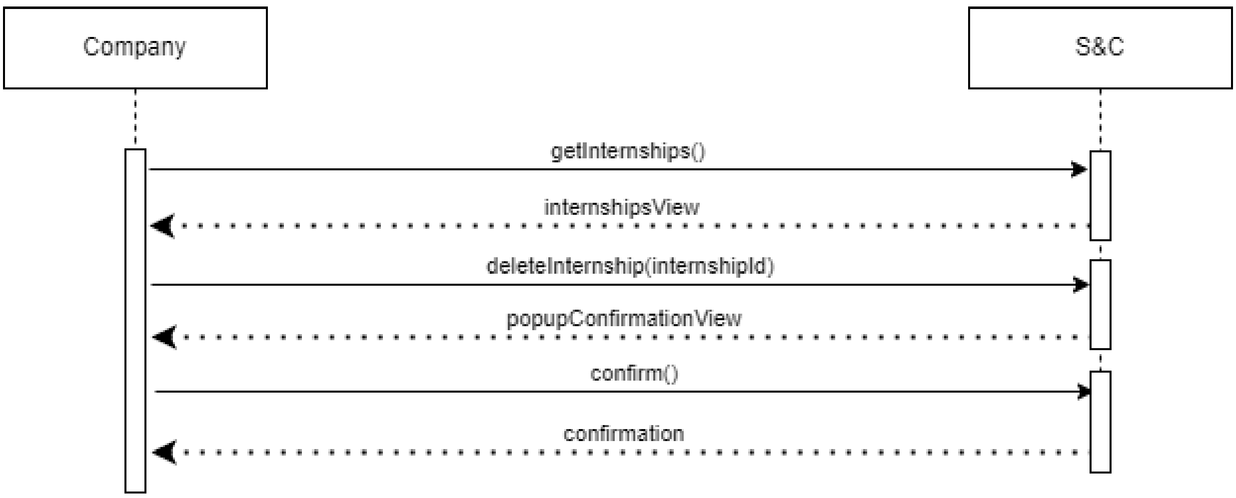
[UC6] Student checks an internship proposed by the system



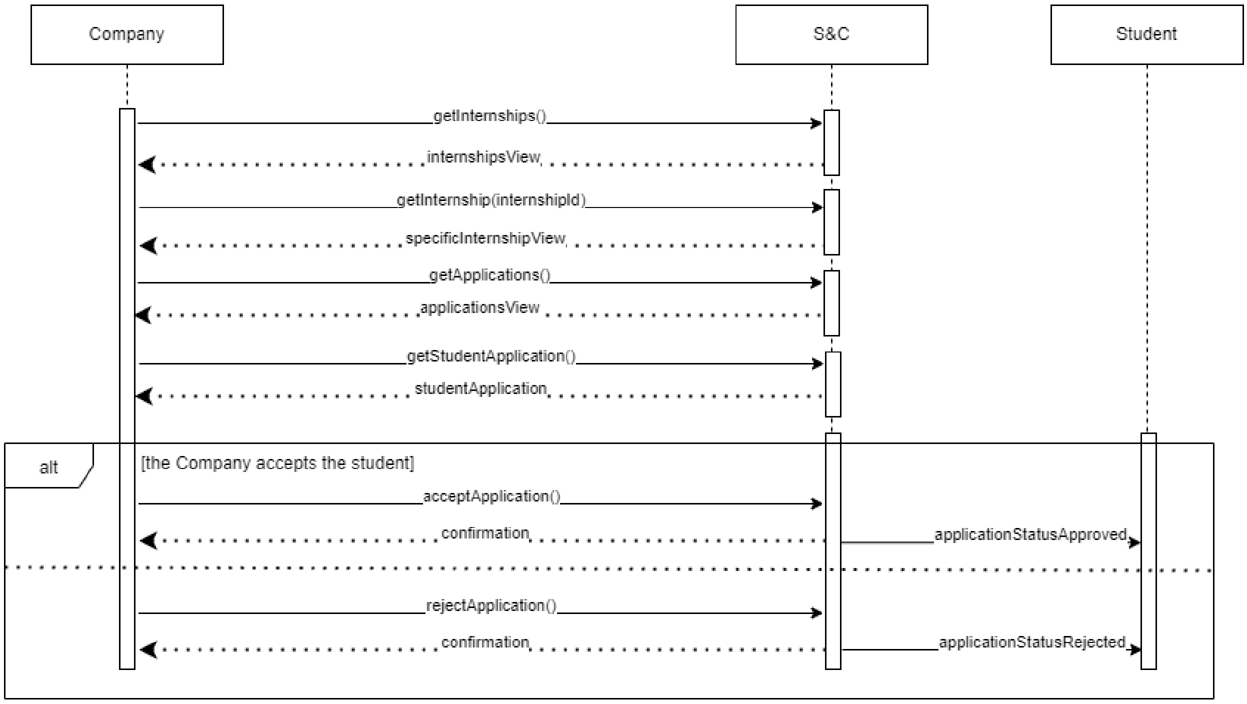
[UC7] Company creates an internship announcement



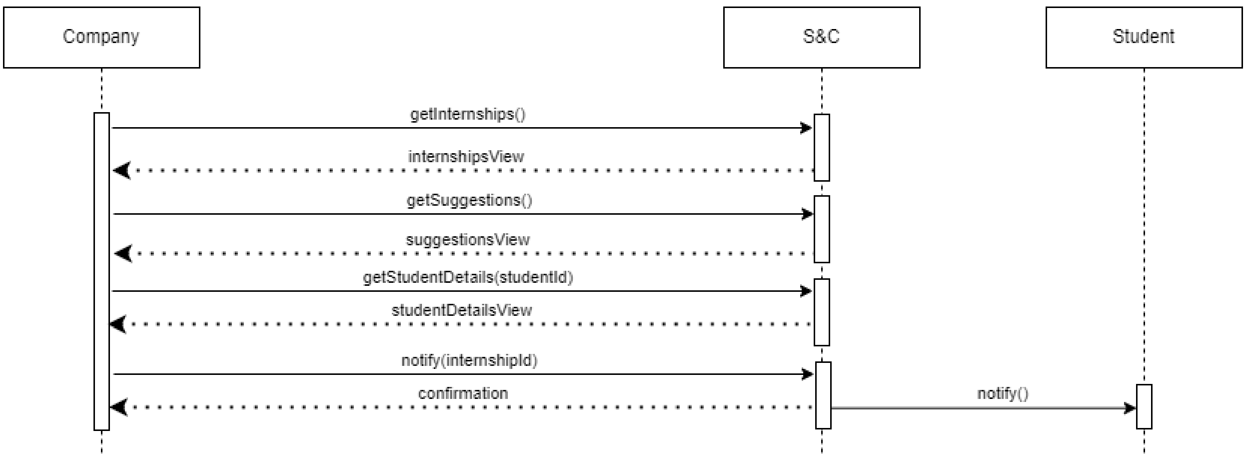
[UC8] Company modifies an internship announcement



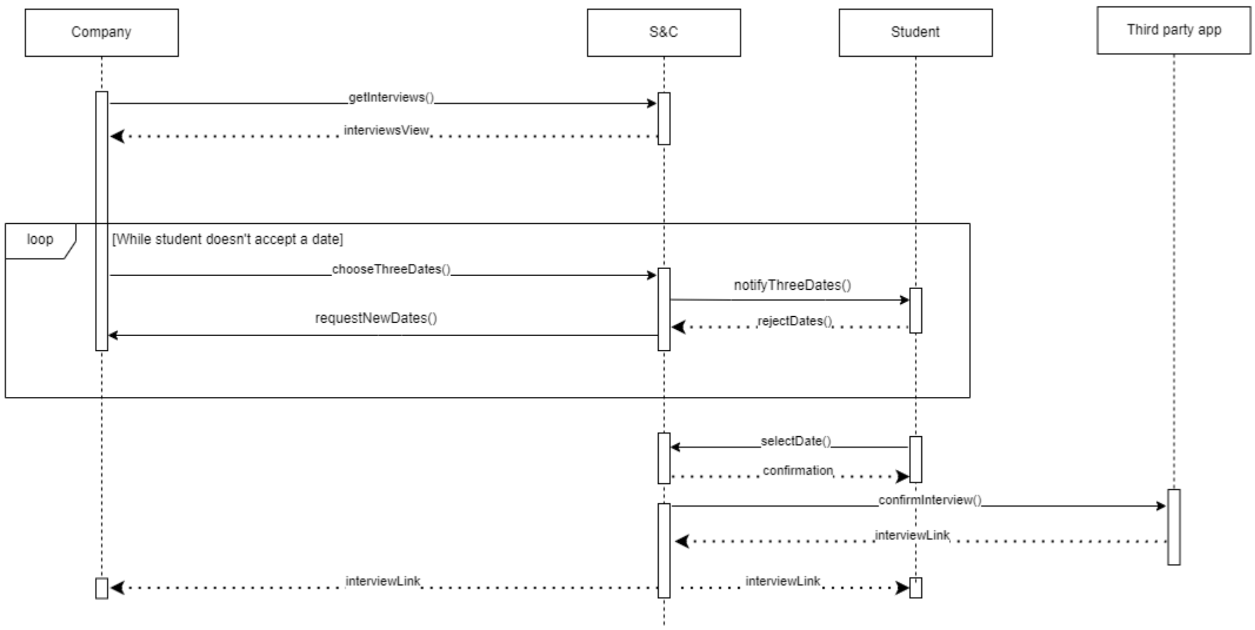
[UC9] Company deletes an internship announcement



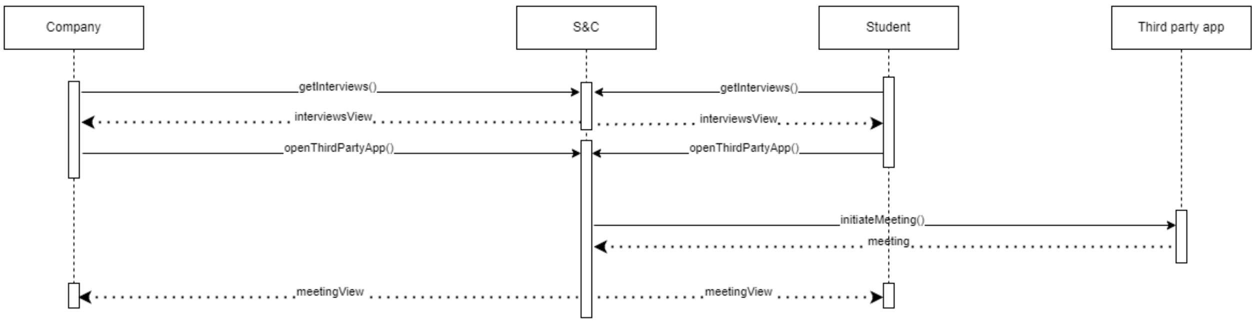
[UC10] Company reviews a student's application



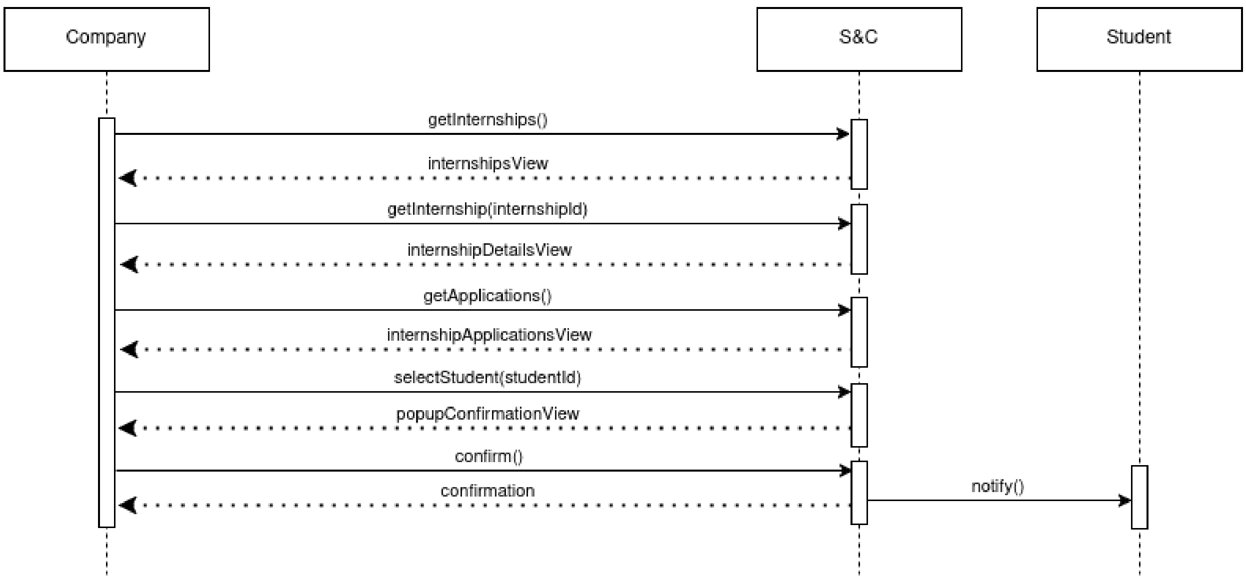
[UC11] Company offers an internship to a student



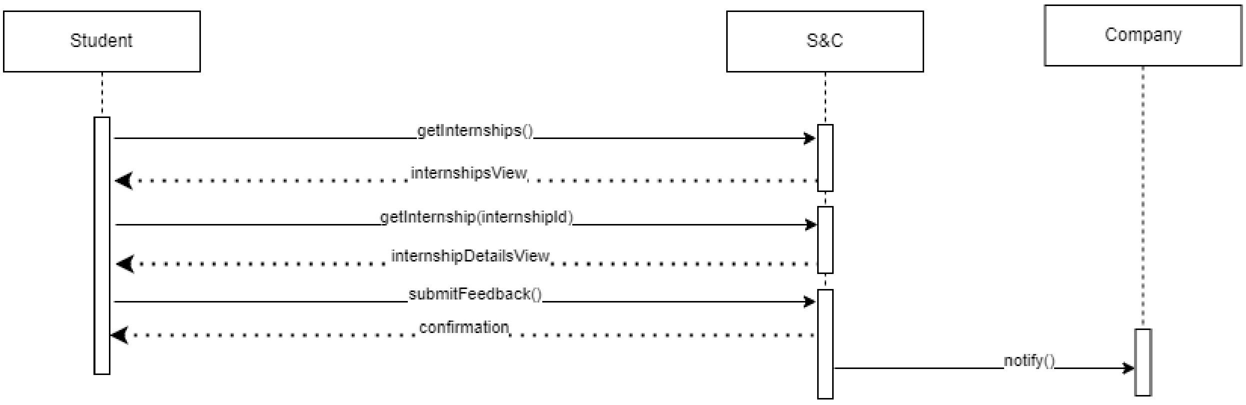
[UC12] Student and company set up the interview



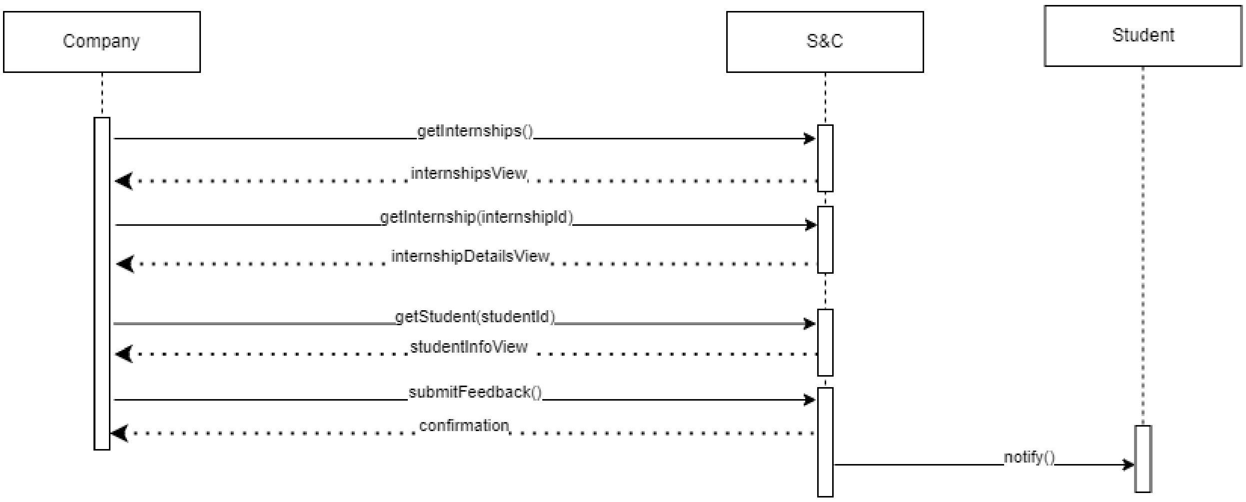
[UC13] Student and company meet for the interview



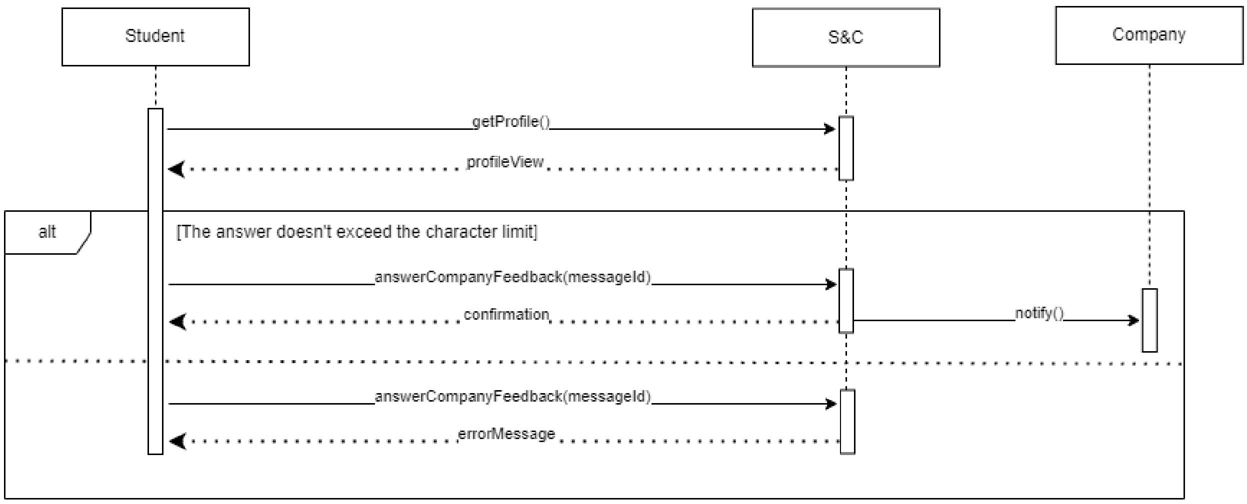
[UC14] Company admits a student to an internship



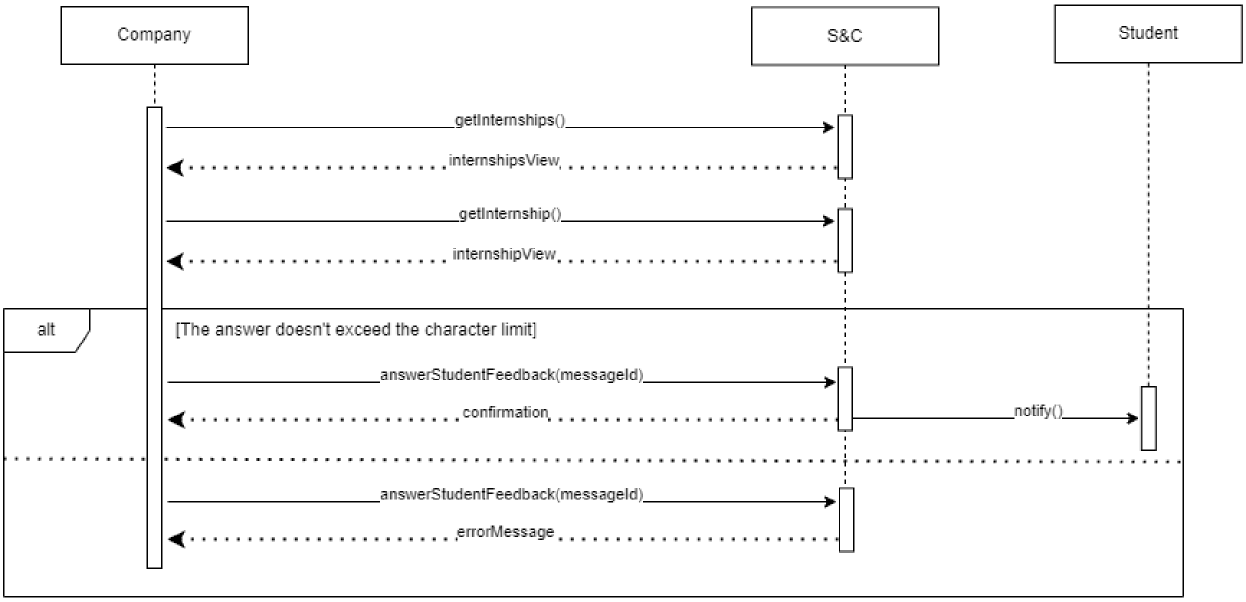
[UC15] Student provides feedback on an internship



[UC16] Company provides feedback on a student



[UC17] Student answers a comment about them



[UC18] Company answers a comment about an internship if offered

3.2.4 Requirements Mapping

Goal ID	**Requirement IDs**	**Domain Assumptions**

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[G1] **Companies find suitable candidates based on the requirements of their projects and positions**	R1 The System shall allow new users to sign up by providing their personal information R2 The System shall allow registered users to log in by providing a username and a password R3 The System shall allow Students to upload a CV R4 The System shall allow Companies to create internship announcement posts containing all the needed details R5 The System shall allow Companies to modify existing internship announcement posts updating the details R6 The System shall allow Companies to delete existing internship announcement posts R7 The System shall allow Students to view the details of any internship R10 The System shall allow Students to accept an internship proposal from a Company R12 The System shall use the Recommendation Engine to analyze student profiles to match them with internships proposed by companies R13 The System shall suggest to Companies Students who match their requirements for an internship proposal R15 The system shall allow companies to review the suggested students to select the ones they wish to contact for an internship proposal R16 The system shall notify Students selected by Companies for an internship	D1 Users have a reliable Internet connection D2 Students provide accurate and up-to-date CVs D3 Companies properly insert information about internships in the announcements

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[G2] **Students find internships matching their profile and experiences**	R1 The System shall allow new users to sign up by providing their personal information R2 The System shall allow registered users to log in by providing a username and a password R3 The System shall allow Students to upload CV R4 The System shall allow Companies to create internship announcement posts containing all the needed details R7 The System shall allow Students to view the details of any internship R8 The System shall allow Students to browse the suggested internships R9 The System shall allow Students to apply for an internship R10 The System shall allow Students to accept an internship proposal from a Company R11 The System shall allow Students to check the status of the applications they've submitted R12 The System shall use the Recommendation Engine to analyze student profiles to match them with internships proposed by companies R14 The System shall suggest to Students Companies that offer internships aligned with their skills and experiences	D1 Users have a reliable Internet connection D2 Students provide accurate and up-to-date CVs D3 Companies properly insert information about internships in the announcements

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[G3] **Students proactively look for internships that align with their skills, experiences, and preferences**	R1 The System shall allow new users to sign up by providing their personal information R2 The System shall allow registered users to log in by providing a username and a password R4 The System shall allow Companies to create internship announcement posts containing all the needed details R7 The System shall allow Students to view the details of any internship R8 The System shall allow Students to browse the suggested internships R9 The System shall allow Students to apply for an internship	D1 Users have a reliable Internet connection D2 Students provide accurate and up-to-date CVs D3 Companies properly insert information about internships in the announcements

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[G4] **Companies are facilitated in managing interviews**	R1 The System shall allow new users to sign up by providing their personal information R2 The System shall allow registered users to log in by providing a username and a password R17 The System shall allow Companies to propose three possible dates to Students for an interview R18 The System shall allow Students to accept one of the proposed dates for an interview R19 The System shall allow Students to ask Companies to provide new possible dates for an interview R20 The System shall allow the hosting of online interviews between Companies and Students via a third party app	D1 Users have a reliable Internet connection D4 Both parties attend the scheduled interviews on the platform
[G5] **Companies manage student admissions to internships**	R1 The System shall allow new users to sign up by providing their personal information R2 The System shall allow registered users to log in by providing a username and a password R21 The System shall allow companies to select students to admit to internships they interviewed for R22 The System shall allow students to check for their admission to internships they interviewed for	D1 Users have a reliable Internet connection D4 Both parties attend the scheduled interviews on the platform D5 Students carry out the internships offered by the companies they were chosen for

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[G6] **Students provide feedback about the internships**	R1 The System shall allow new users to sign up by providing their personal information R2 The System shall allow registered users to log in by providing a username and a password R23 The System shall ask Students to provide feedback about an internship by filling out a form R24 The System shall allow Companies to publicly comment on the feedback they received R25 The System shall allow Students to see feedback about Companies from past internships they offered	D1 Users have a reliable Internet connection D5 Students carry out the internships offered by the companies they were chosen for
[G7] **Companies provide feedback about the internships**	R1 The System shall allow new users to sign up by providing their personal information R2 The System must allow registered users to log in by providing a username and a password R26 The System shall ask Companies to provide feedback about the Student's performance by filling out a form R27 The System shall allow Students to publicly comment on the feedback they received R28 The System shall allow Companies to see feedback about Students from past internships they partook in	D1 Users have a reliable Internet connection D5 Students carry out the internships offered by the companies they were chosen for

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3.3 Performance Requirements

The system has to guarantee good performance to serve many users (companies and students). To achieve this goal the user experience must be as good as possible to fulfill this requirement the response time must be low no more than a second. If the user's internet connection is slow the response time can increase significantly

3.4 Design Constraints

3.4.1 Standards compliance

The S&C platform must handle personal information such as CVs. Because of this, it is designed to meet rigorous standards for data privacy and accessibility. It complies with the EU's General Data Protection Regulation (GDPR), ensuring users' data is handled securely and with full consent. In particular, being based in Italy, it complies with the Personal Data Protection Code, also known as "Codice in materia di protezione dei dati personali" (Legislative Decree No. 196/2003), which implements the GDPR into Italian law. Additionally, the platform follows international best practices for data handling and time formatting, ensuring compatibility with modern systems and providing an efficient global user experience

3.4.2 Hardware limitations

To ensure optimal performance, users must meet certain hardware specifications for running the S&C platform:

- Network Connectivity: a reliable internet connection is required for seamless access to the platform. The device should be compatible with at least one of the following communication standards: 3G, 4G, 5G, Wi-Fi (IEEE 802.11), or Ethernet (IEEE 802.3)
- Device Capabilities: the platform requires a device with at least a quad-core processor (e.g., Intel i5 or equivalent), a minimum of 8 GB RAM, and a screen resolution of at least Full HD (1920x1080). Devices should also support modern web browsers and ensure adequate processing power

In addition to this a webcam is strongly recommended to conduct the interviews on the third party application in a professional and complete manner

3.4.3 Any other constraint

The user interface (UI) of the S&C platform must be intuitive and easy to navigate, given that it serves both students and companies with varying technical proficiency. The design must prioritize accessibility, with clear visual cues and well-structured content. The platform must adapt to different devices, ensuring consistent performance on desktops, laptops, and mobile devices.

3.5 Software System Attributes

3.5.1 Reliability

Reliability is a core attribute for the S&C platform, as it is expected to be available for continuous use. To prevent system downtime, the platform must implement robust failover mechanisms and data replication strategies. This ensures that users can access the platform at all times, even during server maintenance or unexpected failures. Furthermore, regular data backups should be scheduled, ensuring no loss of information

3.5.2 Availability

The platform must achieve a high level of availability. A target of 99.9% uptime should be set, ensuring minimal disruptions. The system architecture should be designed to handle large volumes of concurrent users, especially during peak times. By integrating load balancing and scalable infrastructure, the platform can dynamically allocate resources to meet demand and avoid bottlenecks, maintaining smooth performance even under heavy load

3.5.3 Security

The S&C platform handles sensitive personal data, so security must be a top priority. All user data, including login credentials and student CVs, must be encrypted at rest and in transit. The platform must implement strong access controls, ensuring that only authorized users can access sensitive resources. Regular security audits should be conducted to identify and resolve vulnerabilities proactively

3.5.4 Maintainability

The S&C platform must be designed for long-term maintainability. To facilitate quick updates and smooth debugging, the platform's codebase should be modular, well-documented, and follow industry best practices. A detailed automated testing framework must be implemented, covering critical features and ensuring that new code does not break existing functionality. Maintaining high code quality is vital for minimizing technical debt, allowing new features to be integrated efficiently and securely

3.5.5 Portability

As a web-based platform, S&C must be platform-independent, meaning it should work across a variety of browsers (e.g., Chrome, Firefox, Safari, Edge) and devices (e.g., smartphones, tablets, laptops). The platform's responsive design will ensure that the user experience is consistent and optimized regardless of screen size or device type.

Chapter 4

Formal Analysis Using Alloy

The Alloy model captures the following key aspects of the system's behaviour and rules:

1. Student participation in internships:
    - A student can only become a "participant" in an internship if their application has been reviewed and explicitly approved by the corresponding company
2. Feedback constraints:
    - Students can provide feedback only for internships they actively participated in
    - Companies can provide feedback only about students who took part in internships offered by the company
3. Application outcomes:
    - After applying to an internship, a student receives either an acceptance or rejection decision
4. Feedback interactions:
    - Companies can provide feedback only about students that took part in an internship offered by them
    - Students can provide feedback only about internships they took part in
    - Companies can comment only on the feedback provided by students regarding internships offered by them
    - Students can comment on feedback provided by companies about them during internships they participated in

For simplicity, the model assumes that internships can't be deleted.

```
sig Student {
   var applies: set Application,
   var participates: set Internship,
   var submits: set StudentFeedback,
   var writes: set StudentComment
}

sig Company {
   var offers: set Internship,
   var submits: set CompanyFeedback,
   var writes: set CompanyComment
}

var sig Internship {
   var participants: set Student
}{
   one c: Company | this in c.offers
}

enum Status { APPROVED, REJECTED, PENDING }

var sig Application {
   var status: one Status,
   var refers: one Internship
}{
  one s: Student | this in s.applies
}

var sig StudentFeedback {
   var about: one Internship
}{
   one s: Student | this in s.submits
}

var sig CompanyFeedback {
   var about: one Student
}{
   one c: Company | this in c.submits
}

var sig StudentComment {
   var about: one CompanyFeedback
}{
   one s: Student | this in s.writes
}

var sig CompanyComment {
   var about: one StudentFeedback
}{
   one c: Company | this in c.writes
}
```

```
fact ApplicationsDontChangeReference {
   always ( all a: Application |
         always (a.refers = a.refers'))
}

fact StudentFeedbacksDontChangeSubject {
   always ( all sf: StudentFeedback |
         always (sf.about = sf.about'))
}

fact CompanyFeedbacksDontChangeSubject {
   always ( all cf: CompanyFeedback |
         always (cf.about = cf.about'))
}

fact StudentCommentDontChangeSubject {
   always ( all sc: StudentComment |
         always (sc.about = sc.about'))
}

fact CompanyCommentDontChangeSubject {
   always ( all cc: CompanyComment |
         always (cc.about = cc.about'))
}

// student participates is the opposite of
//internship participants
fact InternshipParticipants {
   always ( all s: Student, i: Internship |
         s in i.participants iff
         i in s.participates)
}

// applications can't be deleted
fact NoApplicationDeletion {
   always ( all a: Application, s: Student |
         a in s.applies implies
         always a in s.applies)
}
```

```
// students remain participants of an internship once approved
fact StudentStayInInternships {
   always ( all s: Student, i: Internship |
         (i in s.participates implies
         always i in s.participates) and
         (s in i.participants implies
         always s in i.participants))
}

// student feedbacks can't change writer or be deleted
fact NoStudentFeedbackDeletion {
   always ( all sf: StudentFeedback, s: Student |
         sf in s.submits implies
         always sf in s.submits)
}

// company feedbacks can't change writer or be deleted
fact NoCompanyFeedbackDeletion {
   always ( all cf: CompanyFeedback, c: Company |
         cf in c.submits implies
         always cf in c.submits)
}

// student comments can't change writer or be deleted
fact NoStudentCommentDeletion {
   always ( all sc: StudentComment, s: Student |
         sc in s.writes implies
         always sc in s.writes)
}

// company comments can't change writer or be deleted
fact NoCompanyCommentDeletion {
   always ( all cc: CompanyComment, c: Company |
         cc in c.writes implies
         always cc in c.writes)
}

// internships can't change owner or be deleted
fact NoInternshipDeletion {
   always ( all i: Internship, c: Company |
         i in c.offers implies
         always i in c.offers)
}
```

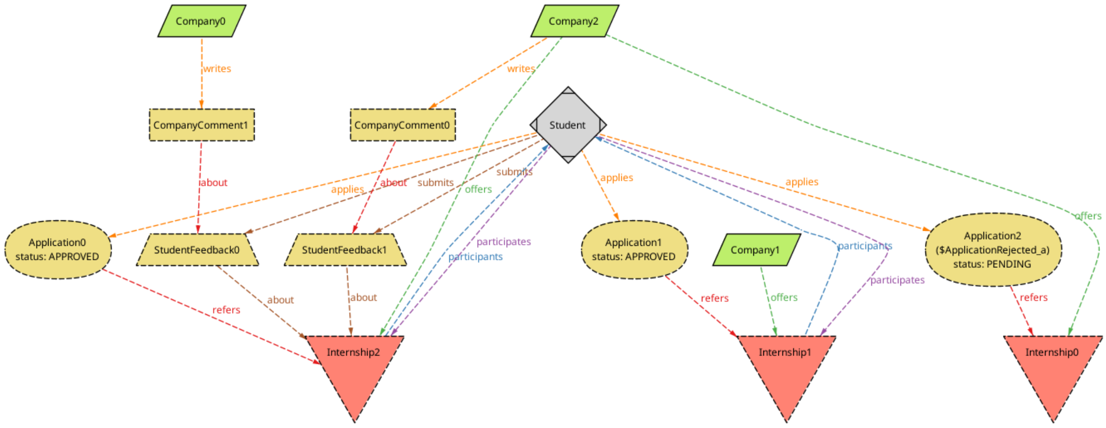
```
// in order to participate in an internship, there must
//have been an application
// and the application must have been approved
fact ParticipationRequiresApprovedApplication {
   always ( all s: Student, i: Internship |
         s in i.participants implies
          some a: Application |
           a in s.applies and
           a.refers = i and
           a.status = APPROVED)
}

// it's not possible to apply twice to the same internship
fact NoDuplicateApplications {
   always ( all s: Student, i: Internship |
         lone a: Application |
          a in s.applies and
          a.refers = i)
}

// different companies offer different internships
fact NoSameInternship {
   always (all c1, c2: Company | c1 != c2 implies
         #(c1.offers & c2.offers) = 0)
}

// students can only provide feedback about
// internships they have participated in
fact StudentFeedbackIfParticipant {
   always ( all f: StudentFeedback, s: Student |
      f in s.submits implies
       s in f.about.participants)
}

// companies can only provide feedback about
// students that participated in one of their internships
fact CompanyFeedbackIfParticipant {
   always ( all f: CompanyFeedback, c: Company |
      f in c.submits implies
       some i: Internship |
        i in c.offers and
        f.about in i.participants)
}

// an approved application can't change its status anymore
fact StatusNotPendingDoesntChange {
   always ( all a: Application |
         (a.status = APPROVED or a.status = REJECTED)
         implies
         always a.status = a.status')
}
```

```
pred ApplicationApproved [a: Application] {
   a.status = PENDING
   a.status' = APPROVED
}

pred ApplicationRejected [a: Application] {
   a.status = PENDING
   a.status' = REJECTED
}
//run ApplicationRejected for 3

pred CompanyCreatesInternship {
   some c: Company | #c.offers' > #c.offers
   some a1, a2: Application | a1 != a2
   after all a: Application | a.status = PENDING
}
//run CompanyCreatesInternship

pred StudentAppliesToInternship {
   some s: Student | #s.applies' > #s.applies and
      some a1, a2: Application |
         a1 in s.applies and a2 in s.applies and
         // application is rejected
         eventually ApplicationRejected[a2] and
         // application approved
         eventually ApplicationApproved[a1] and
         // student is selected for the internship
         eventually s in a1.refers.participants
}
//run StudentAppliesToInternship

pred CompanyWritesFeedback {
   some c: Company | #c.submits' > #c.submits and
      c.offers = c.offers'
}
//run CompanyWritesFeedback for 2

pred StudentWritesFeedback {
   some s: Student | #s.submits' > #s.submits and
      s.applies = s.applies'
}
//run StudentWritesFeedback for 2

pred CompanyWritesComment {
   some c: Company | #c.writes' > #c.writes and
      c.offers = c.offers'
}
//run CompanyWritesComment for 2

pred StudentWritesComment {
   some s: Student | #s.writes' > #s.writes and
      s.applies = s.applies'
}
//run StudentWritesComment for 2
```

```
assert NoStudentLeaveInternship {
   always ( no s: Student | some i: Internship |
         s in i.participants and s not in i.participants')
}
//check NoStudentLeaveInternship for 10 steps

assert ParticipantsIffApproved {
   always ( no s: Student, i: Internship |
      s in i.participants and no a: Application |
         a in s.applies and a.refers = i and a.status = APPROVED)
}
//check ParticipantsIffApproved

assert NoChangeInternshipOwner {
   always (no i: Internship, c: Company |
         i in c.offers and eventually i not in c.offers')
}
//check NoChangeInternshipOwner for 10 steps

assert NoChangeApplicaitonOwner {
   always ( all a: Application | no disj s1, s2: Student |
         a in s1.applies and a in s2.applies')
}
//check NoChangeApplicaitonOwner

assert FeedbackFromStudentParticipant {
   always ( no s: Student, sf: StudentFeedback, i: Internship |
      sf.about = i and sf in s.submits and s not in i.participants)
}
//check FeedbackFromStudentParticipant

assert FeedbackOnStudentParticipant {
   always ( no c: Company, cf: CompanyFeedback, s: Student |
      cf.about = s and cf in c.submits and no i: Internship |
         i in c.offers and s in i.participants)
}
//check FeedbackOnStudentParticipant

assert StudentCommentOnRelativeFeedback {
   always( no sc: StudentComment, cf: CompanyFeedback, c: Company |
      sc.about = cf and cf in c.submits and no i: Internship |
         cf.about in i.participants and i in c.offers)
}
//check StudentCommentOnRelativeFeedback

assert CompanyCommentOnRelativeFeedback {
   always ( no cc: CompanyComment, sf: StudentFeedback, s: Student |
      cc.about = sf and sf in s.submits and
      // sf.about is the internship
      s not in sf.about.participants
}
//check CompanyCommentOnRelativeFeedback
```

4.1 Examples

Application rejected and application approved



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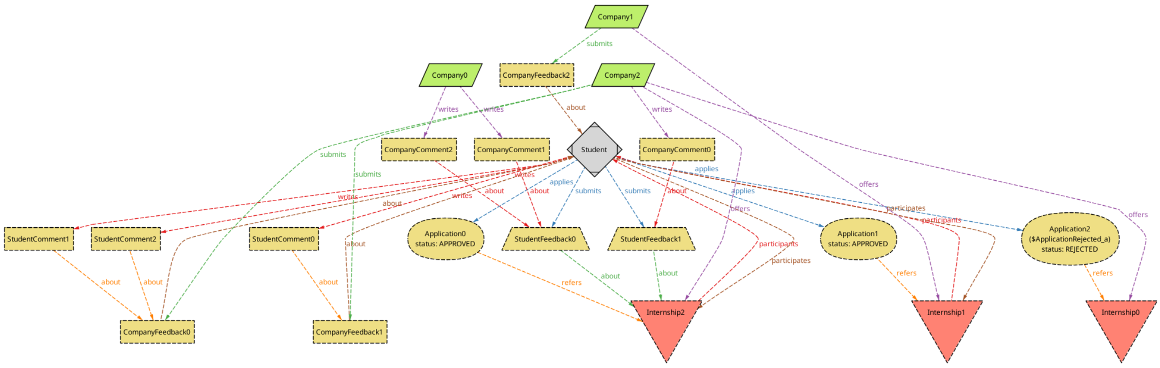

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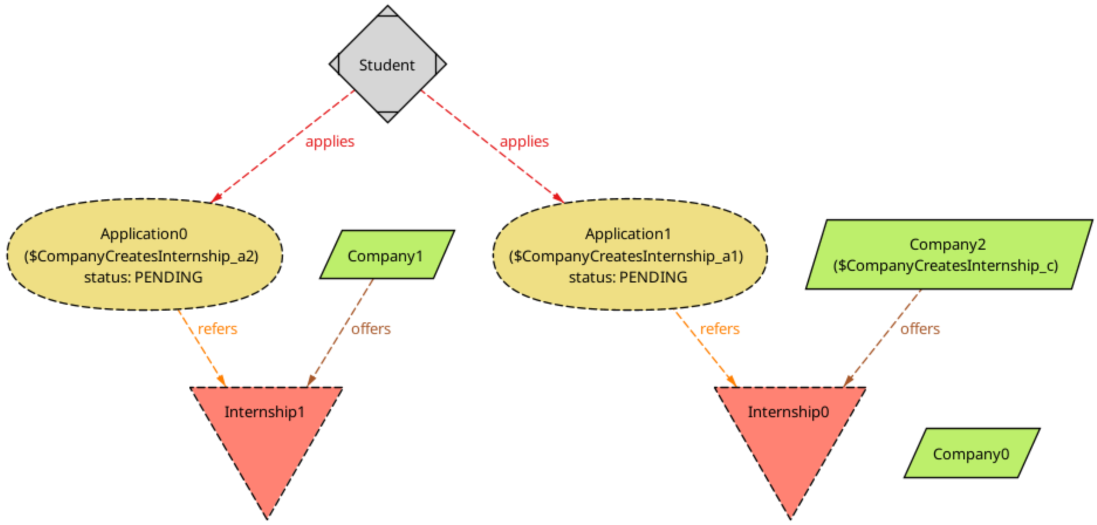
Company creates an internship



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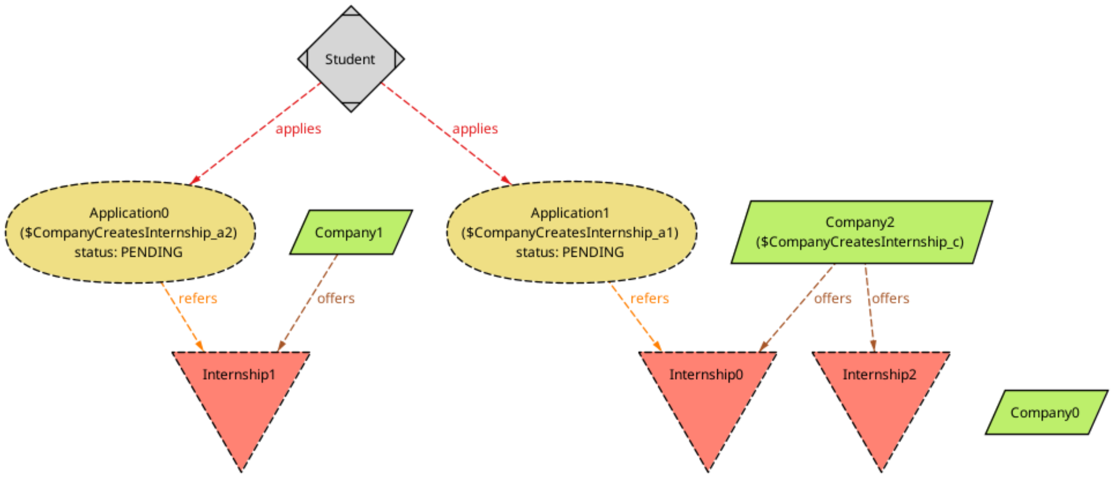

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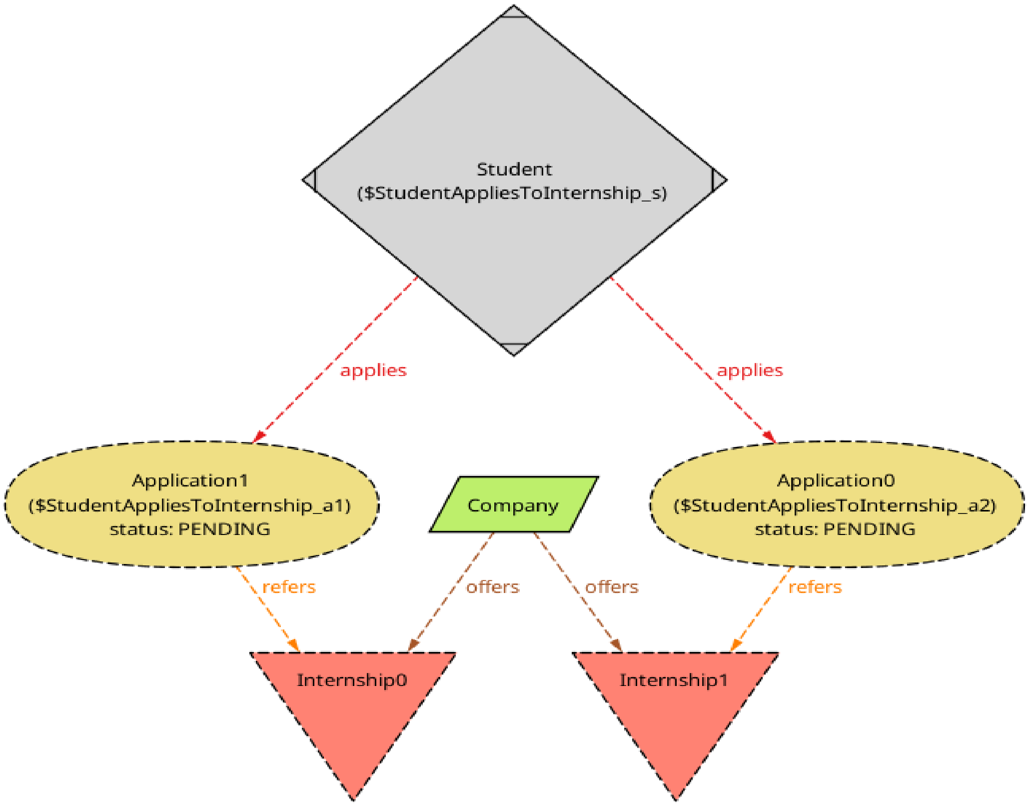
Student applies to an internship



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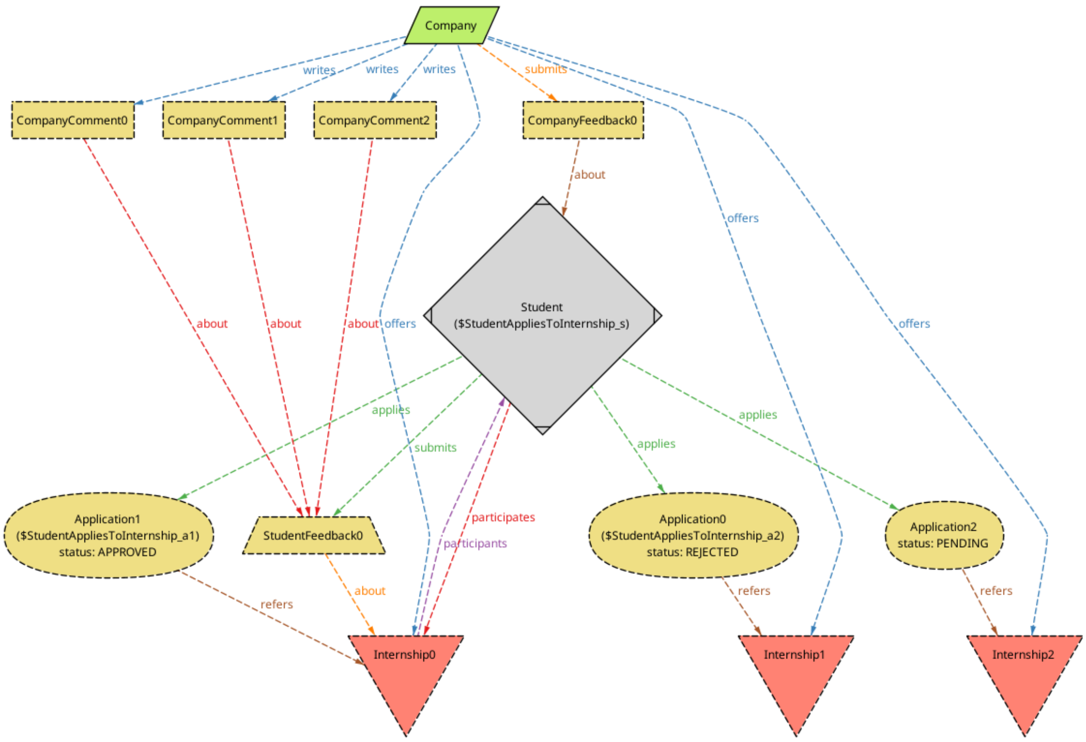
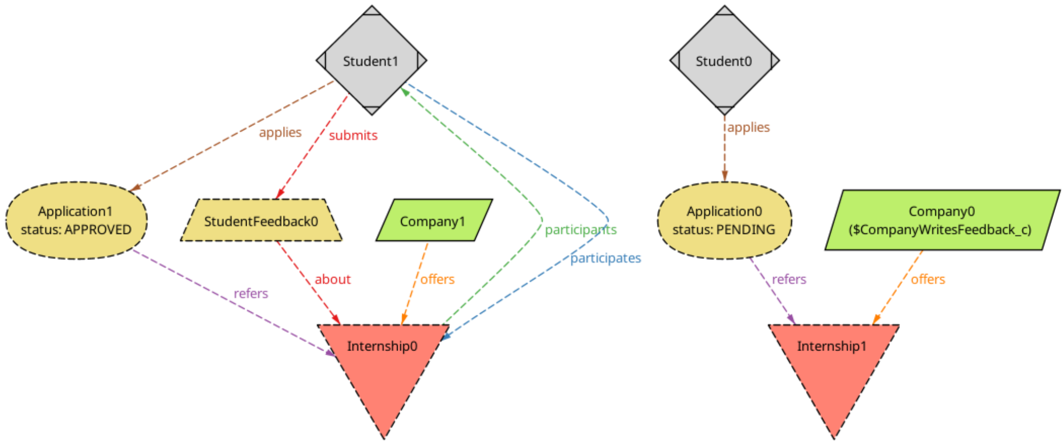


T=1

Company writes a feedback



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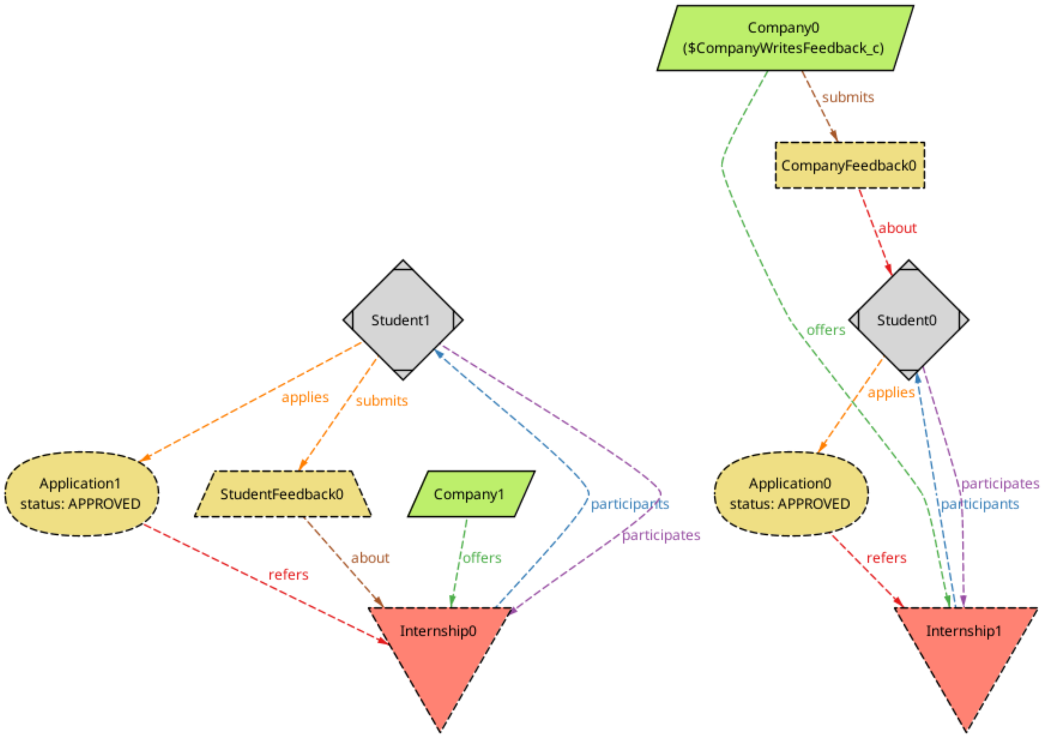
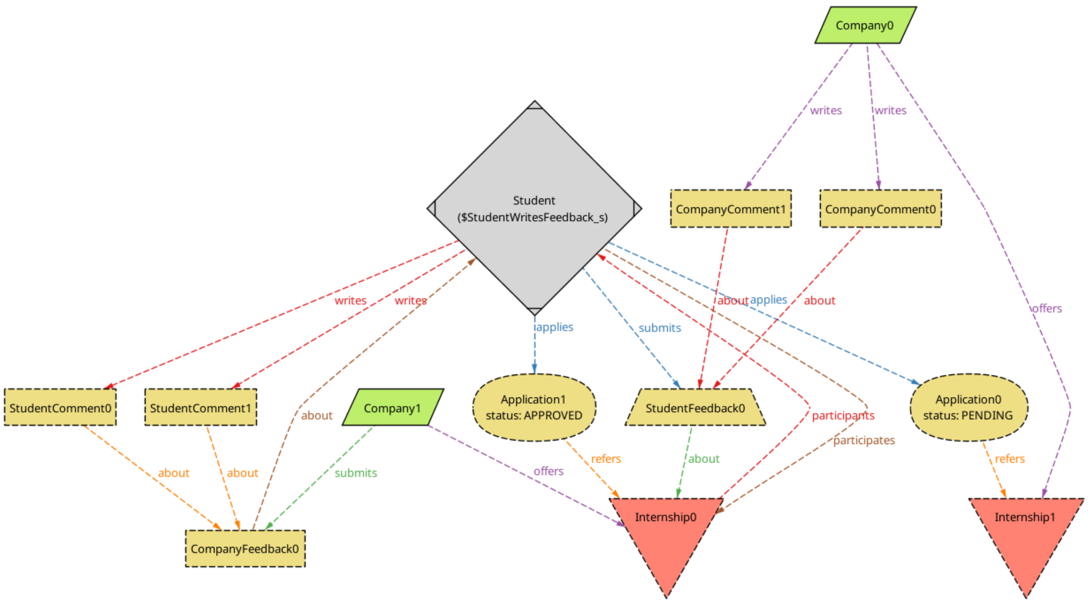


T=1

Student writes a feedback



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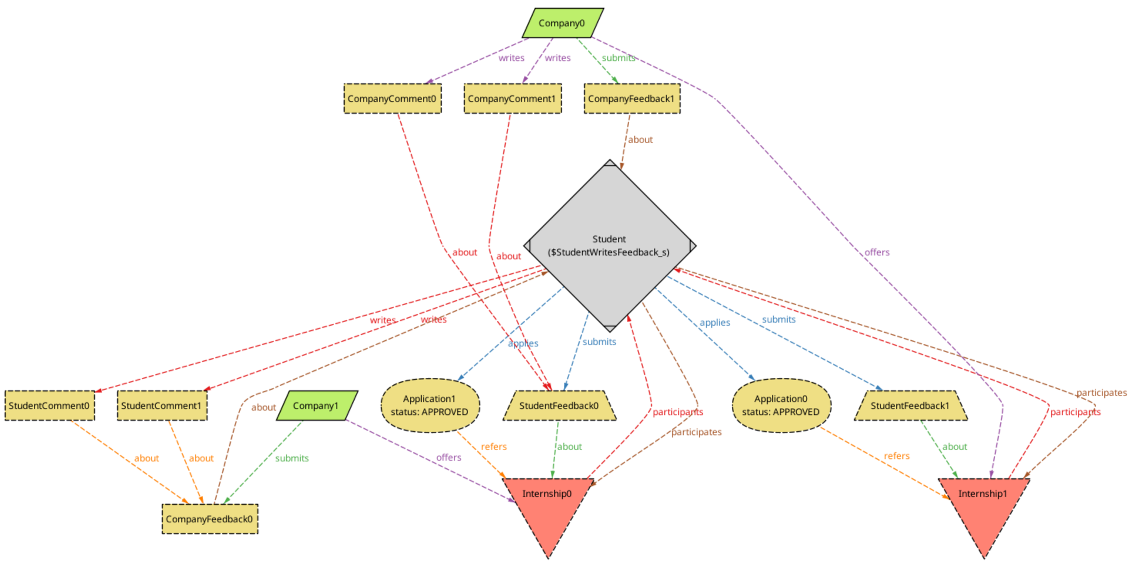
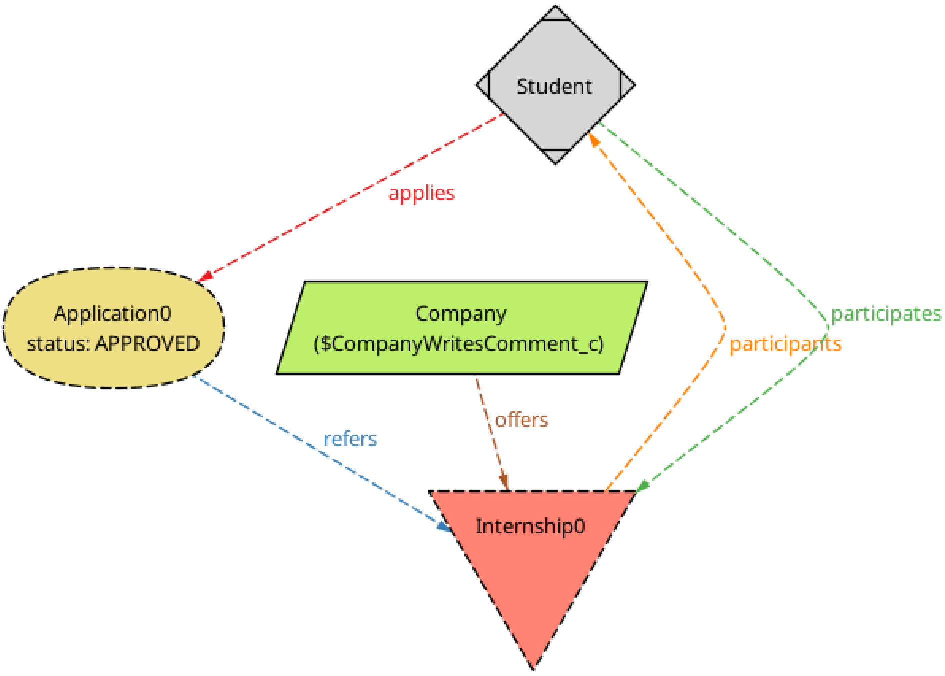


T=1

Company writes a comment



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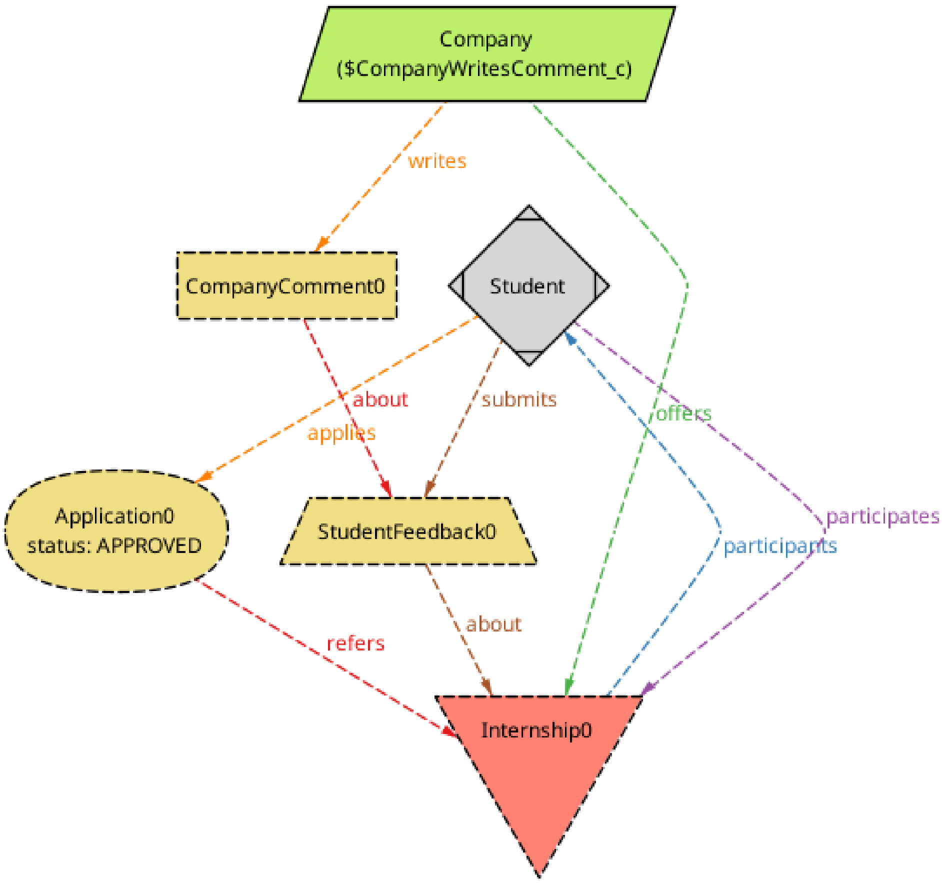
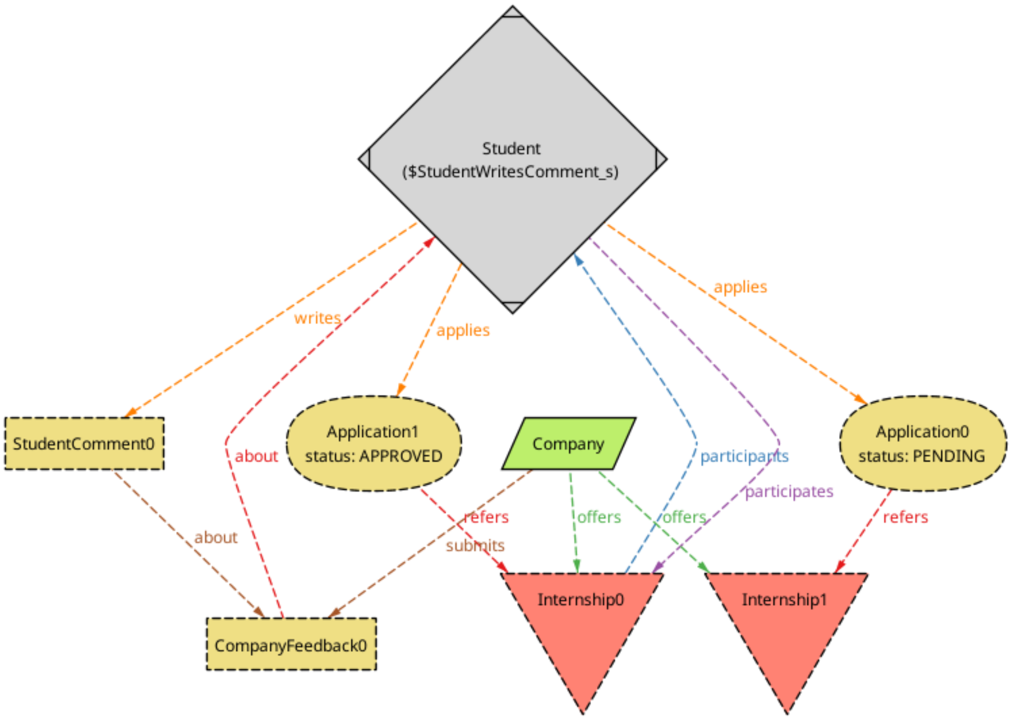


T=1

Student writes a feedback



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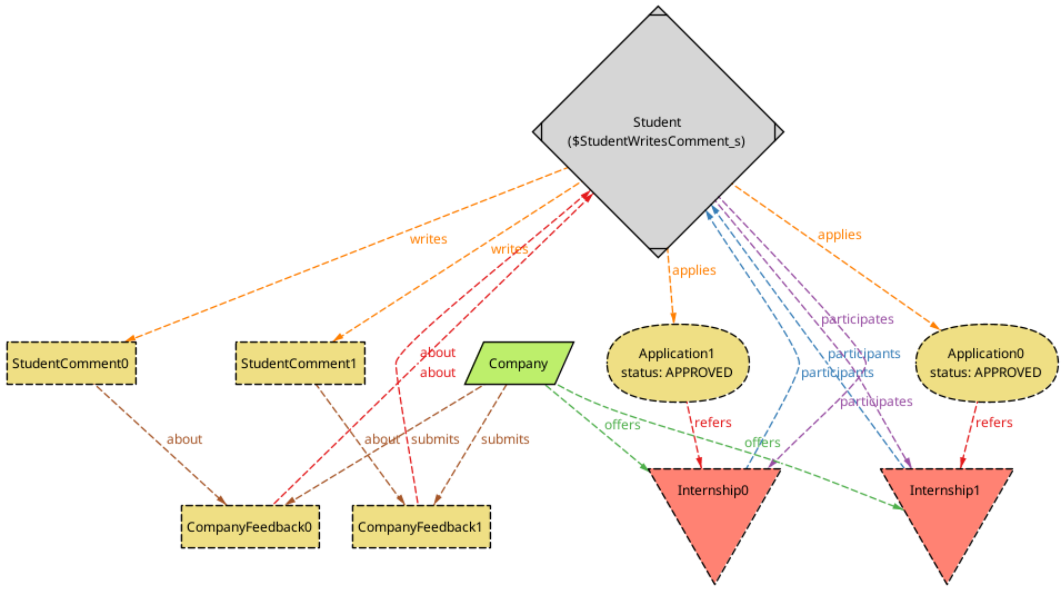

T=1

Chapter 5

Effort Spent

This section contains a record of the effort spent by each group member expressed in hours of work for each section of each chapter

Chapter	**Section**	**Ilardi's hours**	**Luraghi's hours**	**Collaborative hours**
Chapter 1	Section 1.1	0	0	1
	Section 1.2	0	0	2
	Sections 1.3, 1.4, 1.5, 1.6	0	0	0.5
Chapter 2	Section 2.1	2	0	7
	Section 2.2	0	1	0
	Section 2.3	0	1	0
	Section 2.4	0	0	1
Chapter 3	Section 3.1	0	4	1
	Section 3.2	9	2	7
	Sections 3.3, 3.4, 3.5	1	0	1
Chapter 4		0	5	1
Chapter 5		1	0	0
Total		13	13	21.5